

Comparing Mutual Information in Finite Samples: Wald and Expanded Welch–Satterthwaite Inference

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A thesis submitted in partial fulfilment of the requirements for the degree of

Master of Complex Systems

School of Computer Science

The University of Sydney

Australia

2026

Declaration

I certify that this thesis is my own work and that all sources used in its preparation have been acknowledged. Material derived from collaborative work, software, and prior research is identified where it appears. This thesis has not been submitted for another degree except where permitted by the relevant University regulations.

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Abstract

Mutual information measures dependence between two variables, but a difference between two sample estimates need not reflect a difference between their populations. This thesis studies two independent categorical samples and tests whether their population mutual information values are equal, without requiring their complete joint distributions or margins to be the same.

Normal Wald inference divides a bias-corrected difference by a plug-in standard error and uses a normal reference. The proposed Expanded Welch–Satterthwaite procedure keeps that statistic but derives effective degrees of freedom from the sampling variability of the complete mutual information variance functional. A change in one table cell also changes the row and column margins and the pointwise information values; the derivation accounts for all of these effects. The resulting Student reference is a moment-matching approximation, not an exact finite-sample distribution.

The completed experiment compares the two methods on fixed positive-support populations from 2×2 to 8×8 , with additional rectangular, unequal-sample, construction and extreme-sparsity checks. Each of 3,111 unique statistical configurations uses 20,000 paired count-table replicates at a two-sided nominal level of 0.05. False positives, detection, numerical validity and runtime are reported separately for individual regimes. The main null grid includes both identical populations and distinct populations with equal MI; their calibration is also reported separately.

Expanded Welch moves some liberal Wald false-positive rates toward 0.05, but makes conservative regimes more conservative and can be less often valid in sparse samples. Wald is closer to nominal in all 24 main null regimes at sample sizes 500 or 1,000. At common MI 0.02 and sample size 50,000, both methods are near nominal in the tested populations; very small MI remains difficult. Expanded Welch has the same computational order as Wald and costs about 1.75 times as much per scalar call in the measured implementation.

Near independence, the effective degrees of freedom can become very small, making Expanded Welch strongly conservative. A supplementary diagnostic also

shows that changing the reference distribution alone cannot correct all finite-sample errors in the statistic and its standard error.

The evidence supports retaining Wald as the simpler general baseline. Expanded Welch may be informative in selected liberal regimes, provided that valid-result frequency and the complete null behaviour are also reported.

Acknowledgements

I am grateful to my supervisor, Associate Professor Joseph T. Lizier, for guidance throughout this research. I also thank the colleagues, friends, and family members whose support made this work possible.

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Chapter 1

Introduction

Suppose two independent groups are observed on the same categorical variables X and Y . Mutual information (MI) describes the strength of association between those variables in each group. If the estimated MI differs, the difference alone does not tell us whether the populations differ: both estimates vary when new observations are drawn. The question in this thesis is whether the two *population* MI values can plausibly be equal.

This is not the usual test of whether X and Y are independent within one group. The null hypothesis here permits dependence in both groups and permits their complete joint distributions to differ. It specifies only

$$H_0 : I(P) = I(Q), \quad H_1 : I(P) \neq I(Q), \quad (1.1)$$

where P and Q are the two joint distributions. The samples from P and Q are independent of each other; that sampling assumption does not imply independence of X and Y within either population.

1.1 Why the comparison is difficult

MI is a nonlinear function of all cell probabilities and their row and column margins. In a small or uneven table, many cells can be empty in the sample even though their population probabilities are positive. The same data are then used to estimate both the MI difference and its standard error. A first-order Normal Wald test is simple, but its finite-sample rejection rate need not be close to the nominal level.

Welch–Satterthwaite inference offers a possible adjustment. Instead of comparing the standardised difference with a normal reference, it compares the same statistic with a Student reference whose effective degrees of freedom describe uncertainty

in the estimated variance. For MI, that variance is itself a nonlinear function of the full table. Hutcheson's earlier information-theoretic test assigns each variance component its sample size as degrees of freedom; the familiar ordinary-Welch analogue instead uses $n - 1$. Neither assignment accounts for the way a changed cell also changes the margins and pointwise MI values. The method examined here estimates those degrees of freedom from the sensitivity of the complete variance functional.

The resulting reference is an approximation, not a guarantee of better inference. A heavier-tailed reference can reduce excessive false positives, but it can also suppress useful detection when Wald already rejects too rarely. Moreover, one method can fail to produce a valid p-value on samples where the other succeeds. Calibration, detection and valid-result frequency must therefore be considered together.

1.2 Research questions

The thesis asks four related questions. First, how can the sampling uncertainty of the estimated MI variance be turned into an MI-specific Welch–Satterthwaite reference? Second, how does that reference affect false positives, detection and numerical validity relative to Normal Wald? Third, how do table size, margins, baseline dependence, the location of association and unequal sample sizes change the comparison? Fourth, do the observed problems diminish with larger samples, and what additional computation does the correction require?

1.3 Contributions to the field

This thesis contributes to two-sample inference for discrete information measures in four connected ways. The established ingredients are first-order MI inference and Satterthwaite moment matching; the contribution lies in connecting, implementing and testing them for the equal-MI problem.

1. **Mathematical contribution.** The thesis derives the sensitivity of the complete plug-in MI variance to every cell probability. The derivative retains the induced changes in both margins and in the pointwise MI values, providing the quantities needed to estimate MI-specific component degrees of freedom.
2. **Methodological contribution.** The resulting Expanded Welch procedure turns that variance sensitivity into a usable two-sample test. Its relationship to the Hutcheson sample-size assignment, the ordinary $n - 1$ assignment and

Normal Wald is made explicit, so the effect of the complete expansion can be isolated from generic Welch correction in general.

3. **Empirical contribution.** A frozen, reproducible experiment evaluates false positives, detection, valid-result frequency and runtime on fixed populations with known MI. It covers 2×2 through 8×8 square tables, two rectangular shapes, unequal samples and stress regimes, while retaining individual regime results instead of hiding them in an overall average.
4. **Practical contribution.** The evidence establishes the limits of a heavier-tailed reference correction. Expanded Welch can improve a liberal Wald test, but it can also worsen an already conservative test and lose valid results in sparse samples. It therefore should not replace Wald generally. This negative result identifies when reference-tail adjustment is useful and when bias, sparsity or failure of the first-order approximation requires a different solution.

Taken together, these contributions provide both a new MI-specific derivation and a clear decision about its practical value. The thesis does not claim that MI inference, information-theoretic Welch tests or Satterthwaite adjustments are themselves new.

1.4 Scope and organisation

The analysis concerns independent multinomial samples on fixed, aligned categorical alphabets. The principal theory assumes strictly positive cell probabilities and positive first-order MI variances. Exact independence is studied separately as a boundary case, not treated as an ordinary positive-MI regime. The final experiment compares Normal Wald with Expanded Welch at a two-sided nominal level of 0.05. It does not establish performance for paired data, continuous MI estimators or alphabets that grow with sample size.

Chapter 2 supplies the statistical background. Chapters 3 and 4 define the shared statistic and derive the correction. Chapter 5 explains how the populations and samples are generated and how the results are measured. Chapter 6 shows the complete landscape before the focused comparisons. Chapters 7 and 8 discuss the consequences and the resulting recommendation. The appendices provide the detailed algebra, construction and source-to-result record.

Chapter 2

Background and Related Work

This chapter develops the concepts needed to compare MI between two categorical populations. It begins with entropy, pointwise MI, and population MI. It then reviews plug-in estimation, first-order inference, the Welch–Satterthwaite approximation, and prior comparisons of information quantities. The final section identifies the gap addressed by the expanded method.

2.1 Information-theoretic foundations

2.1.1 Entropy and information content

For a discrete random variable X with probability mass function $p_X(x)$, the information content of outcome x is

$$h_X(x) = -\log p_X(x). \quad (2.1)$$

Rare outcomes have larger information content because they are less expected. Using natural logarithms gives information in nats. The Shannon entropy is the expected information content,

$$H(X) = -\sum_x p_X(x) \log p_X(x). \quad (2.2)$$

Entropy summarises the uncertainty of one random variable before its value is observed ([Shannon, 1948](#)).

For a pair (X, Y) , the joint entropy $H(X, Y)$ is defined from the joint probabilities in the same way. The conditional entropy $H(X | Y) = H(X, Y) - H(Y)$ measures the uncertainty remaining in X after observing Y .

2.1.2 Pointwise and average mutual information

Let $P = (p_{ij})$ be the joint distribution of categorical variables $X \in \{1, \dots, r\}$ and $Y \in \{1, \dots, c\}$. Write

$$p_{i+} = \sum_{j=1}^c p_{ij}, \quad p_{+j} = \sum_{i=1}^r p_{ij} \quad (2.3)$$

for the row and column margins. The pointwise mutual information associated with cell (i, j) is

$$\ell_P(i, j) = \log \left(\frac{p_{ij}}{p_{i+}p_{+j}} \right). \quad (2.4)$$

The denominator is the cell probability under independence with the same margins. A positive value means that the cell occurs more often than independence predicts; a negative value means that it occurs less often.

Mutual information is the probability-weighted average of these pointwise values:

$$I(P) = \sum_{i=1}^r \sum_{j=1}^c p_{ij} \ell_P(i, j). \quad (2.5)$$

Equivalently,

$$I(X; Y) = H(X) + H(Y) - H(X, Y). \quad (2.6)$$

The quantity is nonnegative and equals zero exactly when $p_{ij} = p_{i+}p_{+j}$ for every cell, including zero-probability cells.

Equation (2.5) is also the Kullback–Leibler divergence between the joint distribution and the product of its margins. MI therefore measures the overall departure from independence, not one particular shape of association.

This scalar summary deliberately discards much of the information in the joint distribution. Two populations can have the same MI while assigning probability to different cells, having different margins, or exhibiting different association patterns. Equality of MI is therefore weaker than equality of distributions. This distinction is central to the present test: the null fixes one scalar functional while leaving the remaining features of each population unrestricted.

The logarithm base determines the unit but not the inferential problem. If base two is used, every pointwise value and MI value is divided by $\log 2$. The standard error is divided by the same constant, so a standardised statistic is unchanged. This thesis uses natural logarithms and reports MI in nats.

2.2 Estimating discrete mutual information

Suppose a sample of size n produces counts N_{ij} . The empirical cell probabilities are

$$\hat{p}_{ij} = \frac{N_{ij}}{n}. \quad (2.7)$$

Substitution into Equation (2.5) gives the plug-in estimator

$$\hat{I}(P) = \sum_{i,j:\hat{p}_{ij}>0} \hat{p}_{ij} \log \left(\frac{\hat{p}_{ij}}{\hat{p}_{i+}\hat{p}_{+j}} \right). \quad (2.8)$$

The zero-cell convention follows from $\lim_{x \downarrow 0} x \log x = 0$. Empty cells contribute no direct term, but they still affect the empirical margins and the finite-sample behaviour of the estimator.

Under multinomial sampling, the entire table is random subject to its fixed total. Cell counts are dependent because increasing one count leaves fewer observations for the others. The plug-in estimator remains consistent for a fixed alphabet, but its finite-sample distribution depends on the complete probability table. Equal alphabet sizes and sample sizes do not imply equal sampling behaviour when one population has much rarer cells.

The plug-in estimator is upward biased. Finite sampling makes an empirical joint table appear more structured than its population distribution, which increases the estimated departure from independence. For a fixed positive $r \times c$ support, the leading bias is

$$\text{Bias}\{\hat{I}(P)\} = \frac{(r-1)(c-1)}{2n} + O(n^{-2}). \quad (2.9)$$

This result follows from the corresponding leading entropy biases and is closely related to Miller's correction (Miller, 1955); the same histogram-MI term appears in Moddemeijer (1989). More complete analyses show that sparse sampling and unknown support can produce substantial higher-order error beyond Equation (2.9) (Paninski, 2003).

The leading term can be seen from the entropy identity in Equation (2.6). For a discrete variable with m positive categories, the plug-in entropy has leading bias $-(m-1)/(2n)$. Applying this result to $H(X) + H(Y) - H(X, Y)$ gives

$$\text{Bias}\{\hat{I}(P)\} \approx -\frac{r-1}{2n} - \frac{c-1}{2n} + \frac{rc-1}{2n} \quad (2.10)$$

$$= \frac{(r-1)(c-1)}{2n}. \quad (2.11)$$

The correction is positive because estimating the larger joint distribution introduces

more entropy bias than estimating the two margins.

Bias matters directly when comparing two populations. If $I(P) = I(Q)$ but $n_P \neq n_Q$, the uncorrected difference has leading bias

$$\frac{(r-1)(c-1)}{2} \left(\frac{1}{n_P} - \frac{1}{n_Q} \right). \quad (2.12)$$

Removing the leading term is therefore part of the effect estimate, not an optional refinement of its reference distribution.

2.3 First-order inference for nonlinear functionals

MI is a nonlinear function of the cell probabilities. When all population cells are positive and the empirical table is close to its population value, a first-order Taylor expansion replaces this nonlinear function locally by a linear combination of the cell-probability errors. The resulting expansion is often expressed through an influence function. The classical influence-curve and functional delta-method treatments are given by [Hampel \(1974\)](#) and [van der Vaart \(1998, Chapter 20\)](#); [Kandasamy et al. \(2015\)](#) apply related von Mises methods to information functionals.

For MI, the observation-level influence function is centred pointwise MI:

$$\psi_P(i, j) = \ell_P(i, j) - I(P). \quad (2.13)$$

It has population mean zero. Its variance is

$$V(P) = \text{Var}_P\{\psi_P(X, Y)\} = \sum_{i,j} p_{ij} \{\ell_P(i, j) - I(P)\}^2. \quad (2.14)$$

The plug-in MI error is, to first order, the sample mean of these influence values. Consequently,

$$\text{Var}\{\hat{I}(P)\} \approx \frac{V(P)}{n}. \quad (2.15)$$

This leading variance for histogram MI was calculated by [Moddemeijer \(1989\)](#), who noted that the calculation also applies to discrete systems. [Brillinger \(2004\)](#) gives the discrete non-null formula explicitly, attributes that result to Moddemeijer, and states alongside it the $\chi^2/(2n)$ limit at independence. For temporally dependent observation pairs, [Moddemeijer \(1999\)](#) develops an estimator of the same sampling variance using methods that account for dependence between observations. The present work assumes independent observations within each sample and does not claim the first-order MI variance formula as new.

This representation has two roles. First, it makes the central limit theorem available because the leading estimation error is an average of independent, mean-zero contributions. Second, it identifies the observation-level quantity whose variance determines the standard error. The raw categories themselves do not have a numerical variance relevant to MI; their effects on the MI functional are represented by centred pointwise MI.

The expansion is local. It describes sampling fluctuations near a fixed positive-support population and does not assert that the plug-in estimator is normally distributed for every finite table. Its accuracy depends on the sample size relative to the probability structure, not only on the total number of observations.

The central limit theorem then gives a normal approximation when $V(P) > 0$. At exact independence, every pointwise-MI value is zero and $V(P) = 0$. The first-order term disappears, so a second-order Taylor expansion is required. This is why the chi-squared limit for testing independence and the normal limit for comparing positive MI values are not competing descriptions of the same regular case.

2.4 Wald and Welch–Satterthwaite inference

2.4.1 Normal Wald inference

A Wald test estimates a parameter difference, divides it by an estimated standard error, and compares the result with its asymptotic normal reference. For two independent MI estimates, the first-order variance contributions add:

$$\text{Var}\{\hat{I}(P) - \hat{I}(Q)\} \approx \frac{V(P)}{n_P} + \frac{V(Q)}{n_Q}. \quad (2.16)$$

Replacing the population quantities by estimates produces the Normal Wald test developed formally in Chapter 3.

Wald inference is asymptotically valid under regularity conditions, but its finite-sample calibration depends on both the quality of the linear expansion and the stability of the estimated standard error. A normal reference does not separately represent the uncertainty of the two estimated variance contributions.

2.4.2 The Welch–Satterthwaite principle

Welch’s test compares two means when their population variances may differ (Welch, 1947). The denominator contains two independently estimated variance

contributions. Satterthwaite’s approximation represents their sum by a scaled chi-squared variable with matching first two moments ([Satterthwaite, 1946](#)). The resulting effective degrees of freedom are

$$\nu = \frac{(\sum_i k_i s_i^2)^2}{\sum_i (k_i s_i^2)^2 / \nu_i}, \quad (2.17)$$

where s_i^2 are variance estimates, k_i are positive weights, and ν_i are their component degrees of freedom.

The final statistic uses a Student distribution rather than a normal distribution. Small effective degrees of freedom produce heavier tails, reflecting greater uncertainty in the estimated denominator. As the component variance estimates become stable, the effective degrees of freedom increase and the Student reference approaches the normal reference.

For a single scalar restriction, normal and chi-squared Wald notation are equivalent. If T is asymptotically standard normal, then T^2 is asymptotically χ_1^2 . The signed form is retained here because the parameter is the scalar difference $I(P) - I(Q)$, its sign is meaningful, and the Welch refinement replaces the normal reference for T by a Student reference. The $(r - 1)(c - 1)$ -degree chi-squared law used for one-table independence is a different second-order result.

The same principle extends beyond ordinary sample variances. [Pan and Wall \(2002\)](#) account for variability in sandwich variance estimates using approximate Student and F references. [Bell and McCaffrey \(2002\)](#) use Satterthwaite moment matching for regression variance estimates under multi-stage sampling after showing that traditional linearisation estimators can have substantial negative bias. [Kauermann and Carroll \(2001\)](#) examine the variability and efficiency of sandwich covariance estimation and propose an adjustment for the resulting confidence-interval undercoverage. Related adjustments and practical guidance are given by [Fay and Graubard \(2001\)](#) and [Imbens and Kolesár \(2016\)](#). These are close methodological precedents: estimating the variability of a variance estimator and using it in a finite-sample reference are established strategies. The present task is to derive the required inputs for the complete discrete MI functional and assess whether the resulting reference is useful.

The direction of the finite-sample problem is nevertheless different here. The cited regression corrections respond to underestimated uncertainty or anti-conservative coverage, so a larger variance estimate or heavier-tailed reference can move inference in the required direction. Near independence, the MI calculation instead gives $E\hat{V} \approx V + d/n$: the plug-in variance is shifted upward. A heavier-tailed Student

reference then reinforces, rather than offsets, the resulting conservatism.

2.5 Prior MI inference and related comparisons

MI uncertainty also has a broader literature than its leading normal approximation. [Roulston \(1999\)](#) derives entropy and MI error estimates using error propagation and examines them numerically. [Hutter \(2002\)](#) and [Hutter and Zaffalon \(2005\)](#) derive moments of the posterior distribution of MI under a Dirichlet model, including higher-order variance and shape information. Their posterior uncertainty has a different conditioning from the repeated-sampling distribution studied here. Their results establish that higher-order MI uncertainty is not a new topic, but do not directly validate the two-sample Student reference in this thesis.

Hutcheson’s test is a direct information-theoretic Welch predecessor. It compares two Shannon diversity estimates with an estimated variance and a Student reference, using

$$\nu_H = \frac{(s_{H,P}^2 + s_{H,Q}^2)^2}{(s_{H,P}^2)^2/n_P + (s_{H,Q}^2)^2/n_Q} \quad (2.18)$$

as its effective degrees of freedom ([Hutcheson, 1970](#)). Here each of $s_{H,P}^2$ and $s_{H,Q}^2$ is the estimated sampling variance of the corresponding entropy estimator $\widehat{H}$, and each variance component is assigned its sample size, not $n_i - 1$. Applying the same assignment to the two MI variance contributions gives a clear historical baseline. It does not, however, derive how the plug-in MI variance itself fluctuates: MI depends on a joint distribution and both margins, so moving mass into one cell changes pointwise values elsewhere in its row and column. The expanded calculation retains these changes.

The question of comparing MI values also has predecessors. In a segregation application, [Mora and Ruiz-Castillo \(2009\)](#) develop inference for an MI index and discuss pairwise comparisons across groups and periods; the later peer-reviewed account places that MI index within the wider segregation literature ([Mora and Ruiz-Castillo, 2011](#)). They do not derive the variance-sensitivity degrees of freedom used here, but the general aim of comparing MI values is not new. [Guha and Chothia \(2014\)](#) use MI for a different two-sample question: whether two continuous distributions are identical. Their MI is between a pooled observation and its sample label; their null is equality of distributions, not equality of two within-population MI values. For discrete data, [Drake and Guha \(2014\)](#) develop an MI-based k -sample test for the same distributional-equality question.

The exact-independence boundary has likewise been studied separately. [Mari-](#)

[nescu and Balcau \(2025\)](#) use a second-order delta-method argument for MI under independence. More generally, their expansion combines a normal term with a linear combination of chi-squared terms; the normal term vanishes at exact independence. That setting explains why a first-order positive-MI comparison must not be presented as a replacement for a one-table independence test and suggests why second-order effects can remain relevant near that boundary.

Table 2.1 distinguishes the relevant established ideas by the problem each addresses. This is not a claim that no other MI comparison has been proposed. It states which pieces are used here and why the final comparison still needs empirical validation.

Table 2.1: Selected prior work, grouped by the question it addresses.

Topic	Established result and relation to this thesis
MI estimation	Miller (1955) and Paninski (2003) establish bias and sparse-sample limitations of plug-in information estimates.
MI variance	Moddemeijer (1989) and Brillinger (2004) give predecessors for the first-order MI variance, while Moddemeijer (1999) treats dependent observation pairs. This thesis instead studies the sampling uncertainty of the plug-in variance estimate itself in the independent two-sample problem.
Welch principle	Satterthwaite (1946) and Welch (1947) provide moment-matching ideas; Hutcheson (1970) applies the same structure to Shannon diversity using component degrees of freedom n_i .
Other comparisons	Mora and Ruiz-Castillo (2009, 2011) study an MI-based segregation index. They do not derive the variance-sensitivity degrees of freedom used here.
Different MI nulls	Guha and Chothia (2014) and Drake and Guha (2014) test distributional equality; Marinescu and Balcau (2025) study one-table independence. These are not the two-population equal-MI null.
Weak-null tests	Chung and Romano (2013) explain why a raw pooled-label permutation need not be valid when the compared parameters agree but $P \neq Q$; Janssen (1997) gives an earlier studentized permutation result for heterogeneous two-sample nulls.

2.6 Resampling and the weak null

Bootstrap methods can approximate a statistic’s repeated-sampling behaviour by drawing from fitted empirical populations, at greater computational cost than a direct analytic table scan ([Efron, 1979](#)). Sparse empirical support can still be a

problem. A label permutation instead pools the observations and reallocates their group labels; it targets a different symmetry assumption.

Permutation requires a separate distinction. Pooling two samples and permuting their group labels is exact under the strong null $P = Q$, because the observations are exchangeable. Under the weak null $I(P) = I(Q)$, the two complete distributions may differ, so raw label permutation generally targets the wrong null. Studentized permutation can recover asymptotic validity for some weak-null parameter comparisons (Janssen, 1997; Chung and Romano, 2013), but it is not the standard one-table MI shuffle used to test independence.

The final comparison uses Normal Wald and the proposed Expanded Welch method, which share the same effect estimate and standard error. Resampling is relevant future work, but is not an exact finite-sample solution for the unrestricted weak null without additional assumptions.

2.7 Literature synthesis and methodological gap

The literature establishes all of the broad ingredients used in this thesis: plug-in MI estimation and bias correction, first-order MI variance, Wald comparison of smooth functionals, and Welch–Satterthwaite moment matching. It also contains prior comparisons of entropy and MI-related indices.

No single ingredient is presented as new. Shannon established MI; Miller and later authors developed finite-sample bias analysis; Moddemeyer and Brillinger give the first-order MI sampling variance, including a dependent-observation extension; and Welch and Satterthwaite provide the general moment-matching structure. Hutcheson already applies that structure to an information quantity using component degrees of freedom n_i . Mora and Ruiz-Castillo also show that inference about differences in an MI index predates this study. The contribution claimed here is narrower: an explicit observation-level sensitivity for the *estimated MI variance*, its use in a plug-in Welch–Satterthwaite calculation, and a controlled evaluation under distinct equal-MI multinomial populations. The literature review does not establish that no earlier derivation of this exact sensitivity exists, so first priority is not claimed.

The remaining task is specific to MI. Hutcheson’s direct information-theoretic predecessor assigns each component its sample size; an ordinary-Welch analogue uses $n - 1$. Both are fixed assignments. By contrast, established sandwich adjustments derive effective degrees of freedom from the estimated variability of the variance estimator itself. The plug-in MI variance requires its own version of that calculation because it is a nonlinear function of the same empirical joint table used to calculate

MI. A change in one observation modifies the cell probabilities, margins, pointwise-MI values, and their weighted variance.

The expanded method addresses this gap by differentiating the complete MI variance functional. The resulting observation-level sensitivity determines the sampling variance of the variance estimate and, through Satterthwaite moment matching, an MI-specific component degree of freedom. The contribution is this explicit functional calculation and its evaluation, within the established approach to variance uncertainty described above. Chapters [3](#) and [4](#) develop this construction.

Chapter 3

Comparing Mutual Information with Normal Wald

This chapter defines the estimand and statistic shared by the two methods in the final experiment. Normal Wald evaluates it against a standard normal reference; Chapter 4 develops a Student reference with estimated degrees of freedom.

3.1 Data, estimand, and assumptions

Let

$$N^P = (N_{ij}^P), \quad N^Q = (N_{ij}^Q) \quad (3.1)$$

be independent $r \times c$ count tables with totals n_P and n_Q . The row and column categories have the same definitions in both populations. The empirical probabilities are

$$\hat{p}_{ij} = \frac{N_{ij}^P}{n_P}, \quad \hat{q}_{ij} = \frac{N_{ij}^Q}{n_Q}. \quad (3.2)$$

The population parameter is the signed difference

$$\Delta = I(P) - I(Q). \quad (3.3)$$

The principal test is two-sided:

$$H_0 : \Delta = 0 \quad \text{against} \quad H_1 : \Delta \neq 0. \quad (3.4)$$

The first-order construction uses the following assumptions.

Assumption 1 (Sampling). *Observations are independent and identically distributed within each population, and the two population samples are independent.*

Assumption 2 (Aligned fixed alphabets). *The row and column categories are fixed in advance and aligned between the two tables.*

Assumption 3 (Positive support). *Every population cell has positive probability. Empirical zero cells are allowed, but widespread observed support loss is recorded as a diagnostic.*

Assumption 4 (Regular MI). *The first-order influence variances $V(P)$ and $V(Q)$ are positive. This excludes exact independence and other degenerate cases from the regular Student approximation.*

All logarithms are natural. MI differences and standard errors are therefore reported in nats. Changing the logarithm base scales both by the same positive constant and leaves the standardised statistic unchanged.

3.2 Plug-in and bias-corrected MI

For the first table, define empirical pointwise MI by

$$\hat{\ell}_P(i, j) = \log \left(\frac{\hat{p}_{ij}}{\hat{p}_{i+}\hat{p}_{+j}} \right) \quad \text{when } \hat{p}_{ij} > 0. \quad (3.5)$$

The plug-in MI estimate is

$$\hat{I}(P) = \sum_{i,j:\hat{p}_{ij}>0} \hat{p}_{ij} \hat{\ell}_P(i, j). \quad (3.6)$$

The same calculation gives $\hat{I}(Q)$.

Let

$$d = (r - 1)(c - 1). \quad (3.7)$$

Under fixed positive support, the leading bias correction gives

$$\hat{I}_{\text{BC}}(P) = \hat{I}(P) - \frac{d}{2n_P}, \quad \hat{I}_{\text{BC}}(Q) = \hat{I}(Q) - \frac{d}{2n_Q}. \quad (3.8)$$

The estimated effect is

$$\hat{\Delta}_{\text{BC}} = \hat{I}_{\text{BC}}(P) - \hat{I}_{\text{BC}}(Q). \quad (3.9)$$

The correction uses the declared alphabet dimensions, not the observed nonempty dimensions. It may produce a negative corrected MI estimate in a sparse sample. The implementation does not clip that estimate at zero. Clipping would replace the stated bias-corrected estimator by a different nonlinear estimator and would change

its finite-sample bias and sampling distribution. When $n_P = n_Q$, the two leading corrections cancel in the difference in Equation (3.9). When the sample sizes differ, their difference remains and removes the leading unequal-sample bias described in Section 2.2.

3.3 First-order variance of plug-in MI

For population P , define

$$V(P) = \sum_{i,j} p_{ij} \{\ell_P(i, j) - I(P)\}^2. \quad (3.10)$$

This is the population variance of pointwise MI. It governs the sampling variance of the complete plug-in estimator because the first-order MI error is an average of centred pointwise-MI contributions. This first-order variance is established for histogram and discrete MI (Moddemeijer, 1989; Brillinger, 2004); the expanded calculation in Chapter 4 concerns the *sampling uncertainty of its plug-in estimate*.

To see this directly, write one observation as $Z_a^{(P)} = (X_a^{(P)}, Y_a^{(P)})$. A first-order Taylor expansion of MI around the fixed population table P gives

$$\hat{I}(P) - I(P) = \frac{1}{n_P} \sum_{a=1}^{n_P} \{\ell_P(Z_a^{(P)}) - I(P)\} + o_p(n_P^{-1/2}). \quad (3.11)$$

The summands are independent, have mean zero, and have variance $V(P)$. The remainder $o_p(n_P^{-1/2})$ is asymptotically smaller than the leading $n_P^{-1/2}$ sampling fluctuation. Appendix A derives the linear term by expanding MI as a function of the cell probabilities. Therefore,

$$\text{Var}\{\hat{I}(P)\} \approx \text{Var}\left[\frac{1}{n_P} \sum_{a=1}^{n_P} \{\ell_P(Z_a^{(P)}) - I(P)\}\right] \quad (3.12)$$

$$= \frac{1}{n_P^2} \sum_{a=1}^{n_P} V(P) \quad (3.13)$$

$$= \frac{V(P)}{n_P}. \quad (3.14)$$

Subtracting the fixed leading bias term in Equation (3.8) does not change this first-order variance.

The observed table estimates $V(P)$ by

$$\hat{V}(P) = \sum_{i,j:\hat{p}_{ij}>0} \hat{p}_{ij} \{\hat{\ell}_P(i, j) - \hat{I}(P)\}^2. \quad (3.15)$$

Define $\hat{V}(Q)$ analogously.

3.4 Shared standard error and test statistic

Because the two samples are independent, their estimated variance contributions add:

$$\widehat{\text{SE}}^2 = \frac{\hat{V}(P)}{n_P} + \frac{\hat{V}(Q)}{n_Q}. \quad (3.16)$$

The common standardised statistic is

$$T = \frac{\hat{\Delta}_{\text{BC}}}{\widehat{\text{SE}}}. \quad (3.17)$$

The null hypothesis is not imposed when calculating $\hat{\Delta}_{\text{BC}}$, $\widehat{\text{SE}}$ or T : they use the observed tables. Equal population MI enters when interpreting the reference distribution, which is centred at zero under H_0 . In particular, Q 's observed estimate and variance are not set to zero merely because the null is being tested. Equations (3.9)–(3.17) are shared by Normal Wald and Expanded Welch. Their difference lies in the reference distribution, not the effect estimate or plug-in standard error.

3.5 Normal Wald reference

Under the regular first-order asymptotics,

$$T \xrightarrow{d} N(0, 1) \quad \text{under } H_0. \quad (3.18)$$

Here both n_P and n_Q increase while the aligned alphabet and the two positive-support population tables remain fixed. The two-sided Normal Wald p-value is

$$p_{\text{Wald}} = 2\{1 - \Phi(|T|)\}, \quad (3.19)$$

where Φ is the standard normal distribution function. The thesis evaluates this test through p-values and rejection rates. It does not claim confidence-interval coverage, which would require a separate coverage experiment.

The normal reference is a large-sample approximation. Wald estimates the standard error from the tables but applies no degrees-of-freedom adjustment.

3.6 What the expanded method must calculate

The shared statistic supplies the estimated effect and its first-order standard error. The direct Hutcheson-style information-theoretic analogue assigns component degrees of freedom n_P and n_Q ; a familiar ordinary-Welch analogue uses $n_P - 1$ and $n_Q - 1$. Both fixed assignments are retained as explanatory baselines. However, $\hat{V}(P)$ is a nonlinear functional of a table whose margins and pointwise MI values also change with the data. The expanded method instead estimates component degrees of freedom from that full functional.

Satterthwaite moment matching requires the first two moments of each positive variance estimate. To first order,

$$\mathbb{E}\{\hat{V}(P)\} \approx V(P). \quad (3.20)$$

Chapter 4 derives $\text{Var}\{\hat{V}(P)\}$ by calculating how the complete variance functional changes when one observation changes the joint table.

Chapter 4

Expanded Welch–Satterthwaite Method

The expanded method retains the bias-corrected difference, standard error, and test statistic from Chapter 3. It evaluates that statistic against an estimated Student reference rather than the standard normal. This chapter derives the required degrees of freedom from the sampling variability of the complete MI variance estimator.

4.1 Roadmap from the Welch–Satterthwaite equation

For independent variance estimates s_i^2 , positive weights k_i , and component degrees of freedom ν_i , the Welch–Satterthwaite equation is

$$\nu = \frac{(\sum_i k_i s_i^2)^2}{\sum_i (k_i s_i^2)^2 / \nu_i}. \quad (4.1)$$

For the two MI estimates, the variance contributions are

$$k_P s_P^2 = \frac{\hat{V}(P)}{n_P}, \quad k_Q s_Q^2 = \frac{\hat{V}(Q)}{n_Q}. \quad (4.2)$$

These are the two terms in the squared standard error. The remaining quantities are the component degrees of freedom ν_P and ν_Q , which should measure how reliably $\hat{V}(P)$ and $\hat{V}(Q)$ estimate their population counterparts.

The derivation proceeds in four stages. First, Satterthwaite moment matching expresses ν_P in terms of the mean and variance of $\hat{V}(P)$. Second, a one-cell perturbation gives the observation-level sensitivity of $V(P)$. Third, the variability

of these sensitivities gives the sampling variance of $\hat{V}(P)$. Finally, the two estimated component degrees of freedom are combined through Equation (4.1).

4.2 Component degrees of freedom by moment matching

Satterthwaite represents a positive variance estimate by a scaled chi-squared variable. For the first population, write the working model as

$$\hat{V}(P) \sim E\{\hat{V}(P)\} \frac{\chi_{\nu_P}^2}{\nu_P}. \quad (4.3)$$

The symbol $\sim$ denotes a working distributional approximation. The model matches moments; it does not assert that the finite-sample distribution of $\hat{V}(P)$ is exactly chi-squared.

For $U \sim \chi_{\nu_P}^2$,

$$E(U) = \nu_P, \quad \text{Var}(U) = 2\nu_P. \quad (4.4)$$

The scaling in Equation (4.3) therefore matches the mean automatically:

$$E\left[E\{\hat{V}(P)\} \frac{U}{\nu_P}\right] = E\{\hat{V}(P)\}. \quad (4.5)$$

Its variance is

$$\text{Var}\left[E\{\hat{V}(P)\} \frac{U}{\nu_P}\right] = \frac{[E\{\hat{V}(P)\}]^2}{\nu_P^2} \text{Var}(U) \quad (4.6)$$

$$= \frac{2[E\{\hat{V}(P)\}]^2}{\nu_P}. \quad (4.7)$$

Matching this quantity to $\text{Var}\{\hat{V}(P)\}$ gives

$$\boxed{\nu_P = \frac{2[E\{\hat{V}(P)\}]^2}{\text{Var}\{\hat{V}(P)\}}}. \quad (4.8)$$

To first order, $E\{\hat{V}(P)\} \approx V(P)$. The required calculation is therefore the sampling variance of $\hat{V}(P)$.

4.3 A one-cell perturbation of the distribution

The sampling behaviour of a smooth plug-in functional can be obtained by measuring its first-order response to one observation. Fix a cell (x, y) and define the perturbed distribution

$$P_\varepsilon = (1 - \varepsilon)P + \varepsilon\delta_{(x,y)}, \quad (4.9)$$

where $\delta_{(x,y)}$ places probability one on cell (x, y) . At $\varepsilon = 0$, the distribution is P . Increasing ε moves a small amount of probability toward the selected cell while preserving total probability one.

For an arbitrary cell (i, j) ,

$$p_{ij}(\varepsilon) = (1 - \varepsilon)p_{ij} + \varepsilon\mathbf{1}\{i = x, j = y\}. \quad (4.10)$$

The joint and marginal probability derivatives at the original distribution are

$$\left. \frac{dp_{ij}(\varepsilon)}{d\varepsilon} \right|_0 = \mathbf{1}\{i = x, j = y\} - p_{ij}, \quad (4.11)$$

$$\left. \frac{dp_{i+}(\varepsilon)}{d\varepsilon} \right|_0 = \mathbf{1}\{i = x\} - p_{i+}, \quad (4.12)$$

$$\left. \frac{dp_{+j}(\varepsilon)}{d\varepsilon} \right|_0 = \mathbf{1}\{j = y\} - p_{+j}. \quad (4.13)$$

The derivatives of the joint probabilities sum to zero because

$$\sum_{i,j} \{\mathbf{1}(i = x, j = y) - p_{ij}\} = 1 - 1 = 0. \quad (4.14)$$

4.4 Change in pointwise mutual information and MI

Along the perturbation path, pointwise MI is

$$\ell_{P_\varepsilon}(i, j) = \log p_{ij}(\varepsilon) - \log p_{i+}(\varepsilon) - \log p_{+j}(\varepsilon). \quad (4.15)$$

Differentiating the three logarithms and substituting the probability derivatives gives

$$\left. \frac{d}{d\varepsilon} \ell_{P_\varepsilon}(i, j) \right|_0 = \frac{\mathbf{1}\{i = x, j = y\}}{p_{ij}} - \frac{\mathbf{1}\{i = x\}}{p_{i+}} - \frac{\mathbf{1}\{j = y\}}{p_{+j}} + 1. \quad (4.16)$$

MI along the path is

$$I(P_\varepsilon) = \sum_{i,j} p_{ij}(\varepsilon) \ell_{P_\varepsilon}(i, j). \quad (4.17)$$

The product rule separates its derivative into the change in probability weights and the change in pointwise-MI values:

$$\begin{aligned} \left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0 &= \sum_{i,j} \{ \mathbf{1}(i = x, j = y) - p_{ij} \} \ell_P(i, j) \\ &\quad + \sum_{i,j} p_{ij} \left. \frac{d\ell_{P_\varepsilon}(i, j)}{d\varepsilon} \right|_0. \end{aligned} \quad (4.18)$$

The first sum is $\ell_P(x, y) - I(P)$. Substitution of Equation (4.16) into the second sum gives $1 - 1 - 1 + 1 = 0$. Hence

$$\boxed{\left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0 = \ell_P(x, y) - I(P)}. \quad (4.19)$$

This result recovers the centred pointwise-MI influence function used in Chapter 3.

4.5 Change in the MI variance

The target is the first-order change in

$$V(P) = \text{Var}_P\{\ell_P(X, Y)\}. \quad (4.20)$$

Write this variance as

$$V(P) = M_2(P) - I(P)^2, \quad M_2(P) = \text{E}_P\{\ell_P(X, Y)^2\}. \quad (4.21)$$

Define the observation-level variance sensitivity by

$$g_P(x, y) = \left. \frac{dV(P_\varepsilon)}{d\varepsilon} \right|_0. \quad (4.22)$$

Differentiating Equation (4.21) gives

$$g_P(x, y) = \left. \frac{dM_2(P_\varepsilon)}{d\varepsilon} \right|_0 - 2I(P) \left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0. \quad (4.23)$$

It remains to differentiate

$$M_2(P_\varepsilon) = \sum_{i,j} p_{ij}(\varepsilon) \ell_{P_\varepsilon}(i, j)^2. \quad (4.24)$$

The product rule gives

$$\begin{aligned} \left. \frac{dM_2(P_\varepsilon)}{d\varepsilon} \right|_0 &= \sum_{i,j} \left. \frac{dp_{ij}(\varepsilon)}{d\varepsilon} \right|_0 \ell_P(i, j)^2 \\ &\quad + 2 \sum_{i,j} p_{ij} \ell_P(i, j) \left. \frac{d\ell_{P_\varepsilon}(i, j)}{d\varepsilon} \right|_0. \end{aligned} \quad (4.25)$$

The first sum is

$$\ell_P(x, y)^2 - M_2(P). \quad (4.26)$$

For the second sum, define the probability-weighted row and column means

$$\mathbb{E}_P\{\ell_P(X, Y) \mid X = x\} = \frac{\sum_j p_{xj} \ell_P(x, j)}{p_{x+}}, \quad (4.27)$$

$$\mathbb{E}_P\{\ell_P(X, Y) \mid Y = y\} = \frac{\sum_i p_{iy} \ell_P(i, y)}{p_{+y}}. \quad (4.28)$$

Substitution of Equation (4.16) gives

$$\begin{aligned} &\sum_{i,j} p_{ij} \ell_P(i, j) \left. \frac{d\ell_{P_\varepsilon}(i, j)}{d\varepsilon} \right|_0 \\ &= \ell_P(x, y) - \mathbb{E}_P\{\ell_P(X, Y) \mid X = x\} - \mathbb{E}_P\{\ell_P(X, Y) \mid Y = y\} + I(P). \end{aligned} \quad (4.29)$$

Combining Equations (4.19), (4.23), (4.26), and (4.29) yields

$$\begin{aligned} g_P(x, y) &= \{\ell_P(x, y) - I(P)\}^2 - V(P) \\ &\quad + 2 \left[\ell_P(x, y) - \mathbb{E}_P\{\ell_P(X, Y) \mid X = x\} \right. \\ &\quad \left. - \mathbb{E}_P\{\ell_P(X, Y) \mid Y = y\} + I(P) \right]. \end{aligned} \quad (4.30)$$

The first line is the change that would arise from the centred squared pointwise-MI value. The bracket accounts for changes in the pointwise-MI values themselves, including the selected joint probability and both margins.

4.6 Sampling variance of the MI variance estimator

Define the population variability of the observation-level sensitivities by

$$\tau^2(P) = \text{Var}_P\{g_P(X, Y)\}. \quad (4.31)$$

The mean of an influence function is zero. Centring explicitly gives the same variance if numerical substitution produces a small nonzero mean.

Assumption 5 (Regular variance sensitivity). *For the first-order Expanded Welch construction, $\tau^2(P) > 0$ and $\tau^2(Q) > 0$.*

A first-order Taylor expansion of the variance functional around P is

$$\hat{V}(P) - V(P) \approx \frac{1}{n_P} \sum_{a=1}^{n_P} g_P(Z_a^{(P)}). \quad (4.32)$$

The observations are independent, so

$$\text{Var}\{\hat{V}(P)\} \approx \frac{1}{n_P^2} \sum_{a=1}^{n_P} \text{Var}_P\{g_P(Z_a^{(P)})\} \quad (4.33)$$

$$= \frac{\tau^2(P)}{n_P}. \quad (4.34)$$

Substitution into Equation (4.8) gives the population component degree of freedom

$$\boxed{\nu_P \approx \frac{2n_P V(P)^2}{\tau^2(P)}}. \quad (4.35)$$

Equation (4.35) depends on relative, not absolute, uncertainty. Since

$$\frac{\text{Var}\{\hat{V}(P)\}}{V(P)^2} \approx \frac{\tau^2(P)}{n_P V(P)^2}, \quad (4.36)$$

the component degrees of freedom are approximately twice the reciprocal of this relative variance. A stable estimate relative to the size of $V(P)$ has large degrees of freedom; a relatively unstable estimate has smaller degrees of freedom and produces a heavier-tailed final reference. This first-order component formula assumes $\tau^2(P) > 0$. Positive $V(P)$ alone does not guarantee that condition: at a stationary point of the variance functional, its first-order sensitivity can vanish even though MI and $V(P)$ are positive. There, a higher-order analysis of $\hat{V}(P)$ would be needed to describe its finite-sample distribution.

4.7 Table-based component degrees of freedom

The observed table replaces every population quantity in Equation (4.30) by its empirical counterpart. The sums below are over occupied cells, because pointwise MI is undefined at an empirical zero. For each occupied cell,

$$\begin{aligned} \hat{g}_P(i, j) = & \{\hat{\ell}_P(i, j) - \hat{I}(P)\}^2 - \hat{V}(P) \\ & + 2 \left[\hat{\ell}_P(i, j) - \frac{\sum_{j'} \hat{p}_{ij'} \hat{\ell}_P(i, j')}{\hat{p}_{i+}} - \frac{\sum_{i'} \hat{p}_{i'j} \hat{\ell}_P(i', j)}{\hat{p}_{+j}} + \hat{I}(P) \right]. \end{aligned} \quad (4.37)$$

In all weighted sums, an empty cell contributes zero; no logarithm or sensitivity needs to be evaluated there. The empirical mean and variance are

$$\bar{g}_P = \sum_{i,j} \hat{p}_{ij} \hat{g}_P(i, j), \quad (4.38)$$

$$\hat{\tau}^2(P) = \sum_{i,j} \hat{p}_{ij} \{\hat{g}_P(i, j) - \bar{g}_P\}^2. \quad (4.39)$$

The estimated component degree of freedom is

$$\boxed{\hat{\nu}_P = \frac{2n_P \hat{V}(P)^2}{\hat{\tau}^2(P)}}. \quad (4.40)$$

Repeating the same calculation for the second table gives $\hat{\nu}_Q$.

4.8 Combined degrees of freedom and final test

Substituting the two variance contributions and component degrees of freedom into Equation (4.1) gives

$$\boxed{\hat{\nu}_{\text{expanded}} = \frac{\{\hat{V}(P)/n_P + \hat{V}(Q)/n_Q\}^2}{\{\hat{V}(P)/n_P\}^2 / \hat{\nu}_P + \{\hat{V}(Q)/n_Q\}^2 / \hat{\nu}_Q}}. \quad (4.41)$$

The expanded two-sided p-value is

$$p_{\text{expanded}} = 2 \left\{ 1 - F_{t_{\hat{\nu}_{\text{expanded}}}}(|T|) \right\}, \quad (4.42)$$

where F_{t_ν} denotes the cumulative distribution function of a Student t variable with ν degrees of freedom. As with Wald, only test behaviour is evaluated; no confidence-interval coverage claim is made.

The expanded calculation does not change $\widehat{\Delta}_{\text{BC}}$, $\widehat{\text{SE}}$, or T . It changes only the quantile used to evaluate that statistic.

Proposition 1 (Nested rejection on common-valid samples). *If the two calculations use the same finite statistic T and Expanded Welch has positive finite degrees of freedom, its two-sided p -value is at least the Normal Wald p -value. Thus, at a common fixed threshold, every Expanded Welch rejection is also a Wald rejection on samples where both are valid.*

Proof. Write a Student variable as $Z/\sqrt{S}$, with independent $Z \sim N(0, 1)$ and $S \sim \chi_\nu^2/\nu$. For $c > 0$, $f(s) = 2\{1 - \Phi(c\sqrt{s})\}$ has $f''(s) = c\phi(c\sqrt{s})(s^{-3/2} + c^2s^{-1/2})/2 > 0$. Since $\mathbb{E}S = 1$, Jensen's inequality gives $\Pr(|Z/\sqrt{S}| > c) > \Pr(|Z| > c)$. At $c = 0$ both tails equal one. $\square$

The proposition does not classify samples on which one method is invalid. In particular, a correction that only changes the reference cannot raise rejection where Wald is already too conservative.

4.9 Worked count-table example

Consider the two observed 3×3 tables

$$N^P = \begin{pmatrix} 35 & 10 & 5 \\ 10 & 20 & 5 \\ 5 & 5 & 10 \end{pmatrix}, \quad N^Q = \begin{pmatrix} 30 & 5 & 5 \\ 5 & 30 & 5 \\ 5 & 5 & 10 \end{pmatrix}. \quad (4.43)$$

Their totals are $n_P = 105$ and $n_Q = 100$. The plug-in MI estimates are 0.13693 and 0.25848 nats. With $d = (3 - 1)(3 - 1) = 4$, the corrected difference is

$$\widehat{\Delta}_{\text{BC}} = \left(0.13693 - \frac{4}{2(105)}\right) - \left(0.25848 - \frac{4}{2(100)}\right) = -0.12059. \quad (4.44)$$

The empirical influence variances are $\widehat{V}(P) = 0.27189$ and $\widehat{V}(Q) = 0.43371$. Hence

$$\widehat{\text{SE}} = \left(\frac{0.27189}{105} + \frac{0.43371}{100}\right)^{1/2} = 0.08323, \quad T = -1.44898. \quad (4.45)$$

Normal Wald gives $p = 0.14734$.

For Expanded Welch, the table-based sensitivity calculation first gives $\widehat{\tau}^2(P) = 0.93212$ and $\widehat{\tau}^2(Q) = 0.82035$. Substitution into Equation (4.40) gives component

degrees of freedom

$$\hat{\nu}_P = 16.65, \quad \hat{\nu}_Q = 45.86. \quad (4.46)$$

Equation (4.41) combines them into $\hat{\nu}_{\text{expanded}} = 59.03$, giving $p = 0.15264$. Both methods fail to reject at the 5% level for these tables. The example shows that Expanded Welch changes the reference, not the observed statistic.

4.10 Operational algorithm and complexity

Table 4.1 summarises one complete comparison.

Table 4.1: Operational Expanded Welch–Satterthwaite algorithm.

Step	Calculation
1	Validate the two aligned count tables and their declared alphabet dimensions; independence of samples is a design assumption.
2	Compute empirical probabilities, pointwise MI, plug-in MI, and the leading bias-corrected difference.
3	Compute $\hat{V}(P)$, $\hat{V}(Q)$, the shared standard error, and T .
4	Compute the row and column pointwise-MI means needed in $\hat{g}_P$ and $\hat{g}_Q$.
5	Compute $\hat{\tau}^2(P)$, $\hat{\tau}^2(Q)$, and the two component degrees of freedom.
6	Combine the components using Equation (4.41).
7	Return the Student p-value, effective degrees of freedom, and validity diagnostics.

Each empirical probability, pointwise-MI value, row mean, column mean, and variance contribution is calculated by a fixed number of table scans. The time complexity is therefore $O(rc)$, and the working memory is also $O(rc)$. The method is deterministic once the two count tables are given.

4.11 Behaviour near independence

The size of the adjustment has a useful local explanation. For a distribution close to independence, a second-order Taylor expansion gives

$$V(P) \approx 2I(P), \quad \tau^2(P) \approx 4V(P). \quad (4.47)$$

Substitution into the component formula then gives

$$\nu_P = \frac{2n_P V(P)^2}{\tau^2(P)} \approx n_P I(P). \quad (4.48)$$

Thus low MI can produce a small effective degree of freedom and a much heavier-tailed reference. The approximation is local: it assumes fixed table dimension, positive probabilities and sufficiently regular cell counts. Very near independence, quadratic sampling variation can still be important, so a first-order Student adjustment need not improve finite-sample calibration.

4.12 Theoretical status

The influence-function calculations in Equations (4.19) and (4.30) are directional derivatives of the corresponding population functionals. Under fixed positive support, they provide the usual first-order asymptotic linear representations.

The scaled chi-squared model in Equation (4.3) and the final Student reference are working approximations. They match the first two moments of the estimated variance and do not provide an exact finite-sample law. Moreover, $\hat{I}(P)$ and $\hat{V}(P)$ use the same observations; their first-order covariance is generally $E_P[\{\ell_P(X, Y) - I(P)\}g_P(X, Y)]/n_P$, not zero. Moment matching the denominator does not establish independence from the numerator or an exact Student law. Under regular large-sample conditions with positive variance sensitivities, $\hat{\nu}_P$, $\hat{\nu}_Q$, and $\hat{\nu}_{\text{expanded}}$ increase, so the Student reference approaches the same normal limit as Wald inference.

The software first requires matching table shapes, at least two rows and two columns, finite nonnegative integer counts, and more than one observation in each table. Wald is then valid only when the corrected difference and standard error are finite and $\hat{V}(P) + \hat{V}(Q) > 10^{-14}$. Expanded Welch additionally requires positive finite component and combined degrees of freedom. Failed conditions are recorded as invalid calculations; no different test is silently substituted. The validation in Chapters 5 and 6 evaluates how often these conditions occur and whether the finite-sample correction improves calibration.

Chapter 5

Experimental Design

The purpose of the experiment is to compare the complete behaviour of two tests, not just their p-values on convenient tables. Each design point fixes two population distributions in advance. Repeated independent samples from those populations show how often each procedure returns a valid result and how often it rejects. Because the population MI values are known, a rejection at equal MI is a false positive; a rejection at unequal MI is detection.

5.1 A population pair before the design grid

Consider two 2×2 populations whose row and column margins are $(0.7, 0.3)$ for P and $(0.8, 0.2)$ for Q . The construction produces, to the displayed precision,

$$P \simeq \begin{pmatrix} 0.53273 & 0.16727 \\ 0.16727 & 0.13273 \end{pmatrix}, \quad Q \simeq \begin{pmatrix} 0.68175 & 0.11825 \\ 0.11825 & 0.08175 \end{pmatrix}. \quad (5.1)$$

These are *probability* tables, not observed counts. Their target MI values are 0.02 and 0.03 nats, so the true magnitude of the difference is 0.01 nats. Sampling new count tables changes the estimated MI values, but does not move this fixed population pair to a new horizontal-axis location. Calculations use the stored unrounded probabilities; the rounded display in Equation (5.1), determine the samples.

5.2 Constructing fixed joint distributions

For one population, let $a = (a_1, \dots, a_r)$ and $b = (b_1, \dots, b_c)$ be prescribed row and column probabilities. The independence table is $B = ab^T$, with $B_{ij} = a_i b_j$.

Association is introduced by

$$R(t) = ab^T + tH, \quad H\mathbf{1}_c = 0, \quad \mathbf{1}_r^T H = 0. \quad (5.2)$$

The zero row and column sums mean that changing t leaves the margins fixed. For the main construction, H is zero except in the first two rows and columns, where it is

$$H_{1:2,1:2} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}. \quad (5.3)$$

Thus t is added to the two diagonal cells and removed from the other two cells. The changes cancel within each affected row and column. The same matrix specifies a 2×2 test and a localised change in a larger table; only its embedding differs. This follows the minimal four-violation sparse alternative used by [Berrett and Samworth \(2021, Section 3\(b\)\)](#). Here it is adapted to compare two fixed populations: their margins are prescribed separately and the perturbation strength is chosen to attain a specified MI and is not used directly as the horizontal-axis effect.

The largest nonnegative strength before a cell with $H_{ij} < 0$ reaches zero is

$$t_{\max} = \min_{H_{ij} < 0} \frac{a_i b_j}{-H_{ij}}. \quad (5.4)$$

The construction solves for $0 \leq t < t_{\max}$ such that $I\{R(t)\}$ equals the requested target. Targets at or beyond the endpoint are not admitted. This endpoint is specific to the chosen path, not the largest possible MI over every table with those margins. Both constructed tables are checked for strictly positive cells, correct margins and target MI before any sampling. [Appendix B](#) gives the numerical details.

For a dimension of size k , the skewed margin used in the design is

$$m_k(\rho) = \left(\rho, \frac{1-\rho}{k-1}, \dots, \frac{1-\rho}{k-1} \right). \quad (5.5)$$

The three principal profiles are specified separately for the two populations in [Table 5.1](#). For a rectangular $r \times c$ table, the formula is applied with $k = r$ to row margins and $k = c$ to column margins. At a zero MI difference, the same-margin profiles produce $P = Q$ under this construction; the different-skew profile has $I(P) = I(Q)$ but $P \neq Q$.

In the usual effect direction, $I(P) = I_0$ and $I(Q) = I_0 + |\Delta|$. The paired solve chooses separate strengths for P and Q ; it does not add random noise to their true MI values. The reverse-direction imbalance experiment assigns the higher target to P . The statistical convention remains $\Delta = I(P) - I(Q)$, which is negative in the

Table 5.1: Row and column margins used for each population in the principal design. “Uniform” means the vector with every entry equal to $1/k$.

Profile	P : rows and columns	Q : rows and columns
Uniform	uniform	uniform
Same skew	$m_k(0.8)$	$m_k(0.8)$
Different skew	$m_k(0.7)$	$m_k(0.8)$

usual direction. Figures use the nonnegative magnitude $|\Delta|$.

Two focused checks change the pattern or constructor while keeping the interpretation of target MI. A “rare block” puts Equation (5.3) in the final two rows and columns. A “spread” pattern is the outer product of equally spaced, centred row and column scores; its row and column sums are zero. The ordinal log-linear comparator fits the prescribed margins while varying an ordinal interaction strength to meet the target MI. Matching margins and MI across constructors does not imply matching all cell probabilities or expected counts.

5.3 Sampling, tests and measured outcomes

For each unique configuration, 20,000 independent pairs of multinomial count tables are drawn from its fixed P and Q . Normal Wald and Expanded Welch are applied to the *same* count-table pair. This pairing makes method differences more precisely measurable than unrelated simulations, though it does not remove Monte Carlo uncertainty. The two-sided nominal threshold is $\alpha = 0.05$ throughout the statistical study.

Let $A_m = 1$ if method m returns a valid p-value no larger than 0.05, and $A_m = 0$ otherwise. The primary plotted rejection rate is

$$\hat{r}_m = \frac{1}{20,000} \sum_{s=1}^{20,000} A_{m,s}. \quad (5.6)$$

At $|\Delta| = 0$ this estimates the false-positive rate. At $|\Delta| > 0$ it estimates the probability of detecting the specified difference at the fixed nominal threshold, often called power. A rate near 0.05 is desirable under the null; a smaller rate is not automatically better. Under an alternative, higher detection is useful only alongside acceptable null calibration and a usable test result.

An invalid calculation contributes no rejection to Equation (5.6). This convention reports the behaviour of the complete procedure, but an invalid result is not a

valid acceptance of the null. The companion valid-result rate is the fraction of replicates with a usable p-value. The saved data also give rejection conditional on each method’s valid outputs and on the subset where both are valid. Those denominators answer different questions: conditioning each method on its own valid samples can change which tables are compared. For a single rate, saved Wilson 95% intervals (Wilson, 1927) represent Monte Carlo uncertainty at the fixed population pair. Paired method differences use the same replicates and their saved paired standard errors. These intervals do not describe uncertainty over all possible population distributions.

The “minimum expected count” diagnostic is $\min_{ij} n_P p_{ij}$ for P , or $\min_{ij} n_Q q_{ij}$ for Q . It is not the expectation of the smallest observed count. No lower bound on that diagnostic was used to admit statistical configurations. Empty observed cells and rows remain possible. In the plots a hollow marker warns that fewer than 90% of outputs of that method were valid; 90% is a display convention, not a validated safety threshold.

5.4 Experiment families and computation

Table 5.2 records the question answered by each family. The common population construction above applies unless the named factor changes. Exact parameter sets are in Appendix B, and the machine-readable protocol is identified in Appendix E. There are 534 distinct population pairs (P, Q) , each specifying two fixed joint probability tables. Reusing a pair at different sample sizes creates distinct sampling configurations. There are 3,111 unique statistical configurations but 4,001 display points: some configurations are deliberately shown in more than one comparison. Each unique configuration is sampled once, giving 62,220,000 table pairs in the final statistical run. The within-family counts in Table 5.2 do not sum to 3,111 because the same configuration can occur in several families. Display reuse is not an independent replication.

The runtime experiment measures a full method call after count tables are already available. It varies table shape, profile, sample size and null or alternative status. Each regime uses 200 saved inputs, 20 warm-up calls and five timed passes. Thus it measures method computation, not the cost of constructing populations or generating random tables. Median call times and interquartile ranges are reported in Chapter 6. The finite tested range includes samples of size one in the stress check and an 8×8 largest square table; it does not imply valid inference for all sparse distributions or unbounded alphabet sizes.

Table 5.2: The fixed experiment families. All statistical points use 20,000 paired replicates and the 5% threshold.

Family	Displays	Unique	Factor examined
Main	648	648	Square table size, three margin profiles and equal sample size.
Baseline	648	648	Starting positive MI of 0.0001, 0.001 or 0.02 nats.
Changed cells	144	144	First, rare or spread association patterns.
Imbalance	1,152	1,056	Sample allocation and which population has higher MI.
Larger effects	162	162	MI differences up to 0.2 nats where feasible.
Rectangular	108	108	2×3 and 3×5 tables.
Construction	432	432	Additive versus ordinal log-linear population tables.
Extreme skew	315	315	Dominant margins up to 0.9995, including $n = 1$ stress.
Rare-cell stress	140	140	First- versus rare-block changes, including $n = 1$ stress.
Independence	144	144	Exact $I(P) = 0$ boundary.
Convergence	108	108	Fixed equal-MI populations with sample sizes up to 50,000.
Runtime	60	60	Complete-call timing regimes, separate from simulation.

Chapter 6

Experimental Results

Every ordinary panel in this chapter begins at a true MI difference of zero. That point measures false-positive rejection under H_0 . Points to its right measure detection under H_1 . The horizontal coordinate is a *constructed population* difference in nats; random count tables create the vertical variation. The blue solid circles are Normal Wald and the magenta dashed squares are Expanded Welch. The dotted horizontal line marks the nominal 0.05 rejection rate. Each plotted rate divides the number of valid rejections by all 20,000 replicates. A hollow marker indicates a valid-result rate below 0.90, not an accepted test. The figure source manifest gives the exact saved configuration identifiers for every panel.

6.1 Main landscape: shape, margins and sample size

The first question is broad: do the methods behave differently as the table grows, margins become uneven and samples become larger? Figures 6.1–6.12 provide a systematic view, not a selection of favourable examples. Each of the four shapes is shown under all three margin profiles. Within every figure, the columns are the same three equal sample sizes. The full vertical scale preserves comparability of detection; the lower row enlarges the region in which calibration differences near 0.05 can be seen. The complete nine-size ladder, including $n = 2, 5, 20, 50, 250, 500$, remains in the companion atlas.

The landscape does not yield a uniformly preferable method. At $n = 5$, Expanded Welch rejects none of the 12 main null regimes. This includes the uniform 3×3 , 5×5 and 8×8 regimes, where more than 90% of calculations are valid, so the zero rates cannot be attributed only to invalid outputs. Among the 24 main null regimes

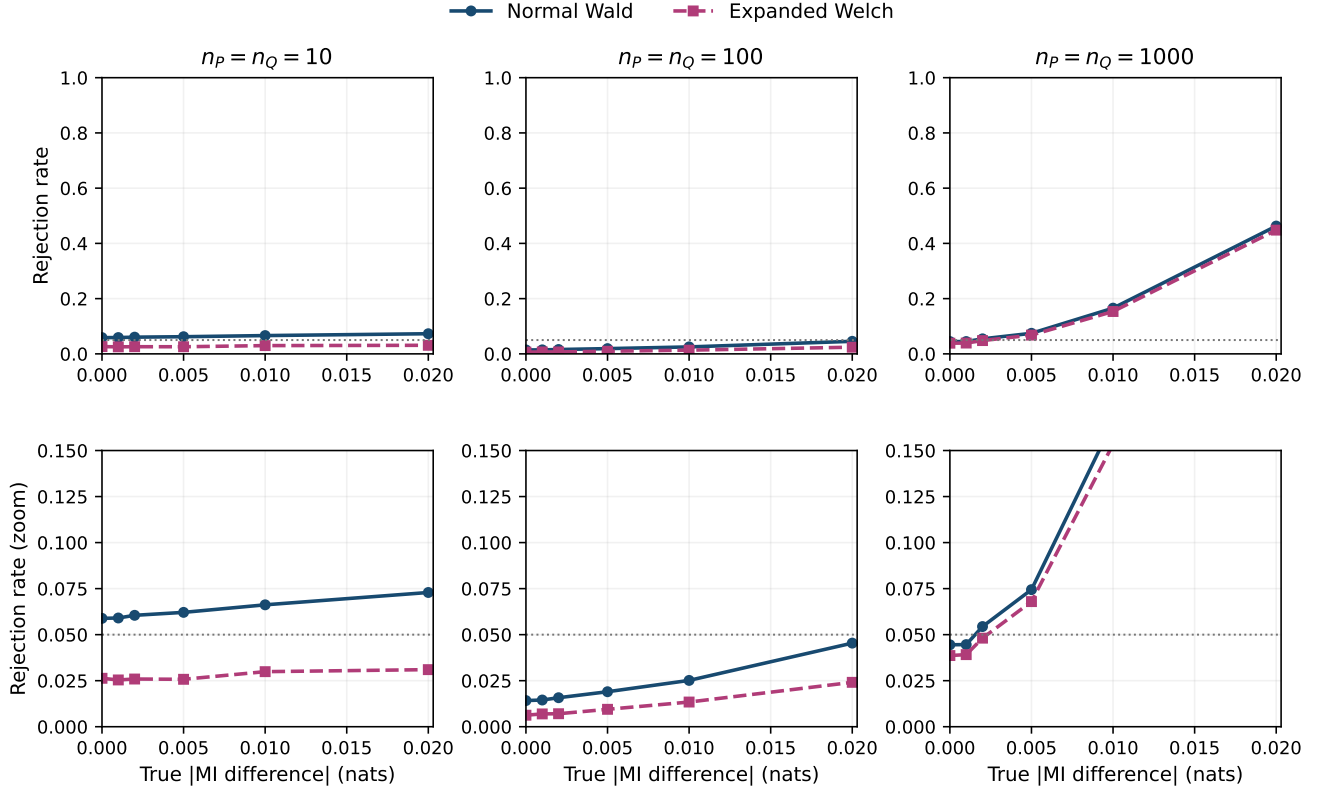


Figure 6.1: 2×2 tables, both P and Q have uniform row and column margins $(1/2, 1/2)$. Columns: $n_P = n_Q = 10, 100, 1000$, respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3); $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The horizontal grid is $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

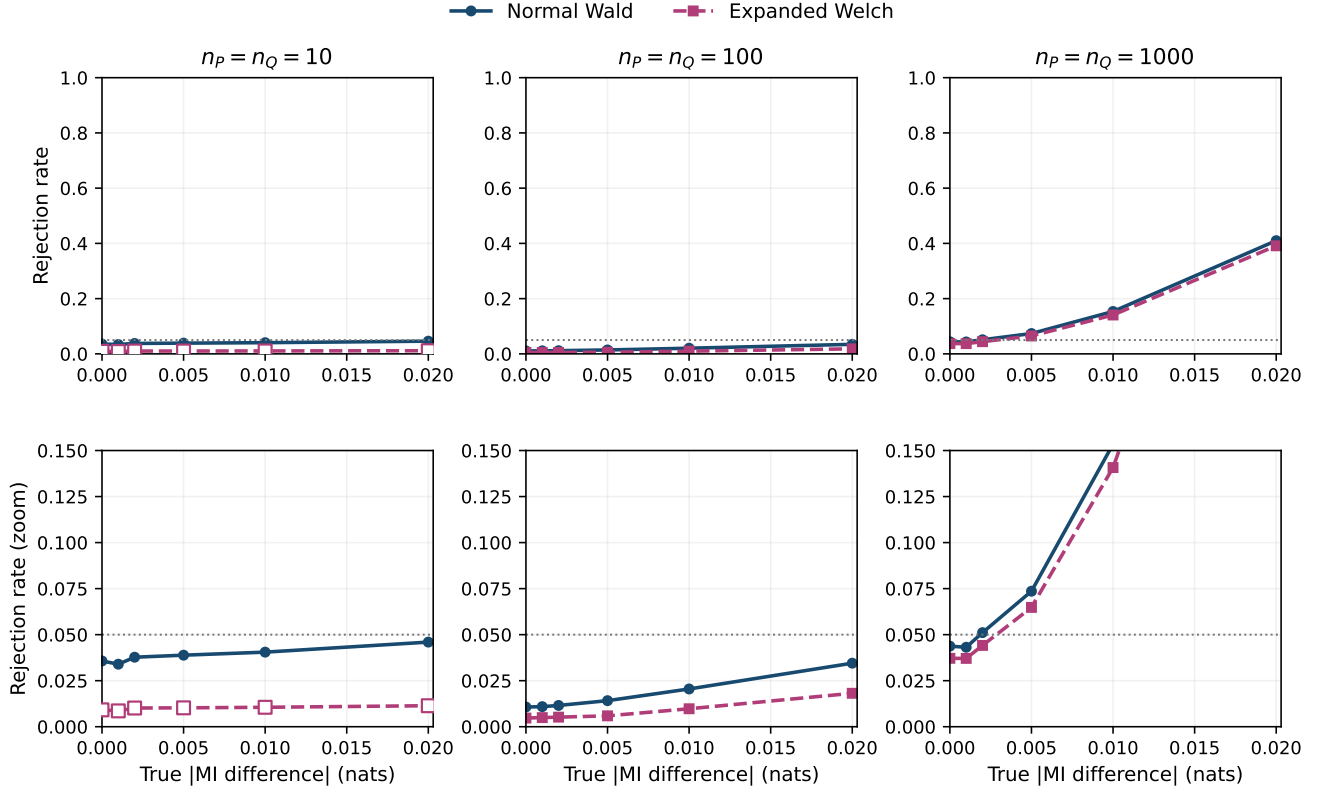


Figure 6.2: 2×2 tables, both P and Q have row and column margins $m_2(0.8) = (0.8, 0.2)$. Columns: $n_P = n_Q = 10, 100, 1000$, respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3); $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The horizontal grid is $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

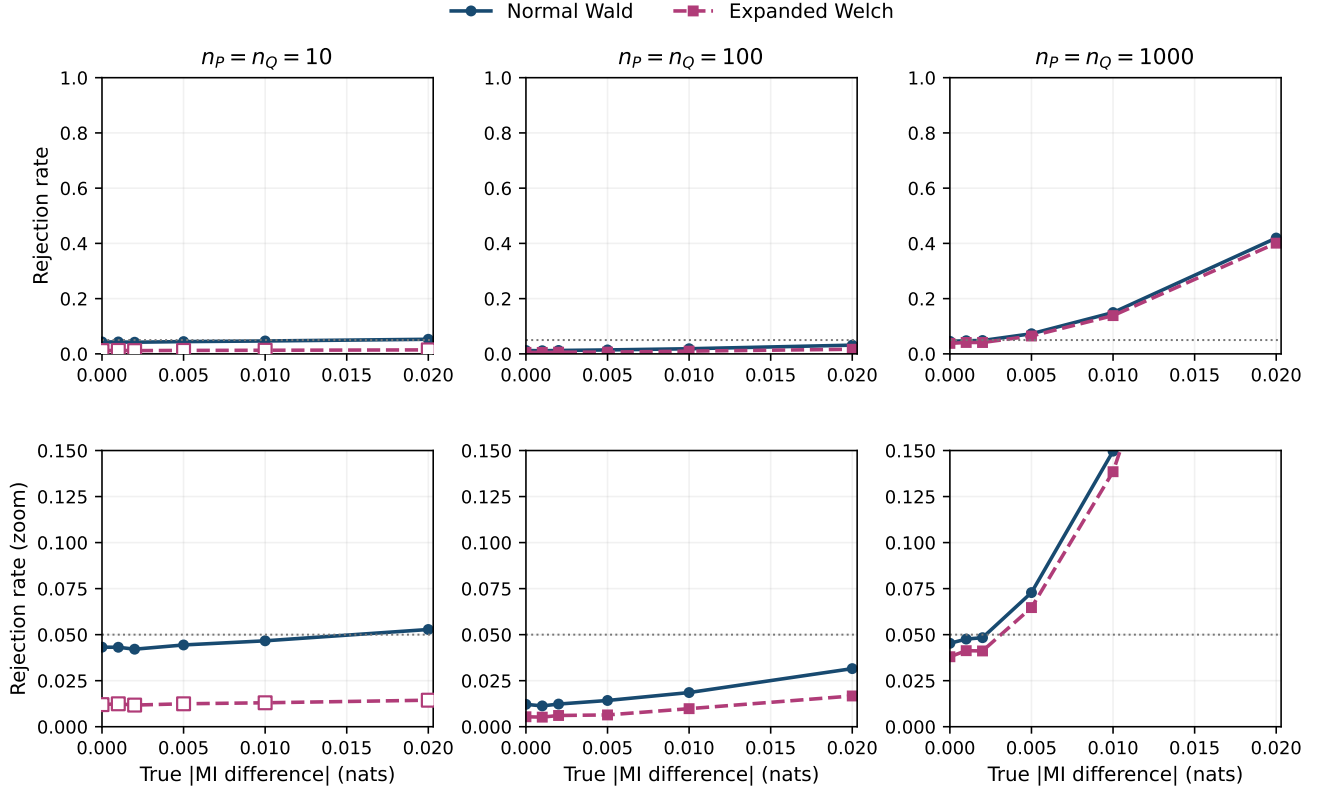


Figure 6.3: 2×2 tables, P has row and column margins $m_2(0.7) = (0.7, 0.3)$, while Q has $m_2(0.8) = (0.8, 0.2)$. Columns: $n_P = n_Q = 10, 100, 1000$, respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3); $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The horizontal grid is $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

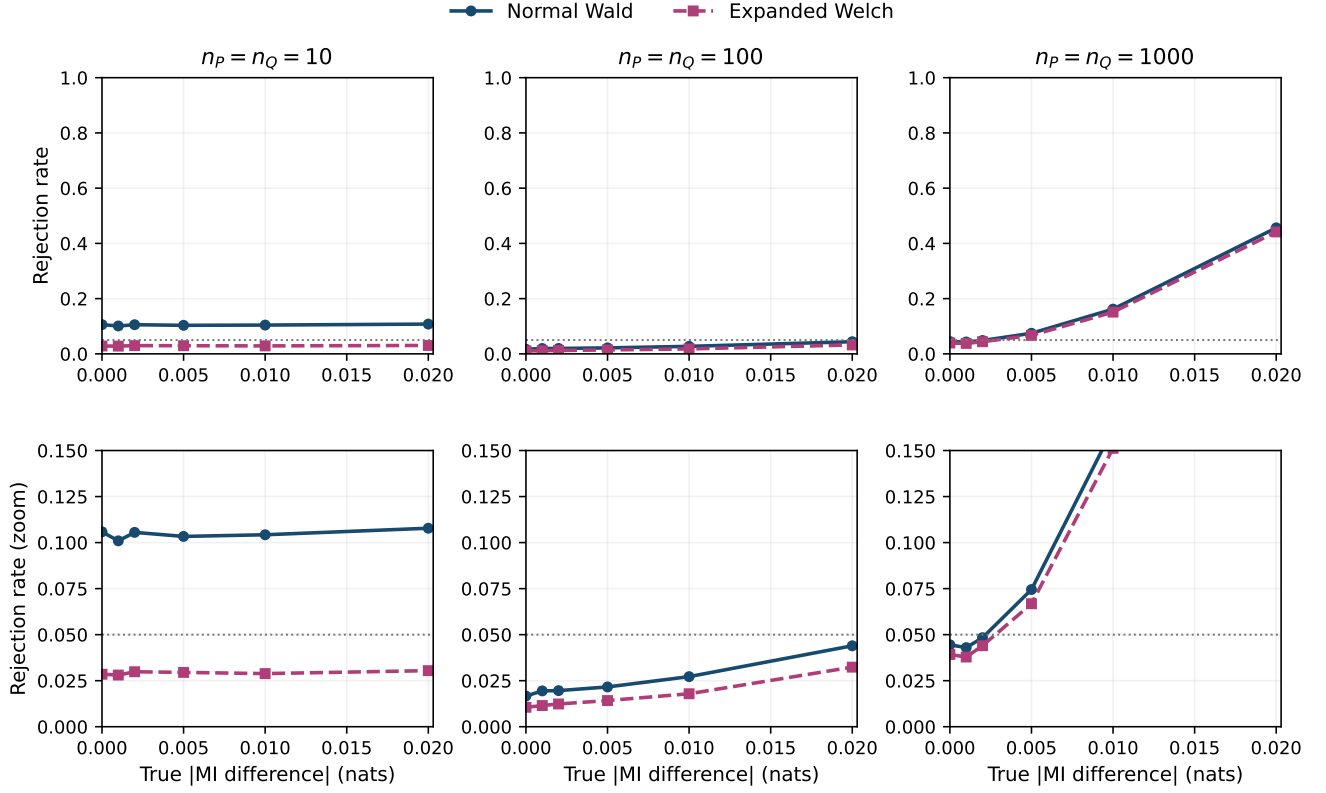


Figure 6.4: 3×3 tables, both P and Q have uniform row and column margins $(1/3, 1/3, 1/3)$. Columns: $n_P = n_Q = 10, 100, 1000$, respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3); $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The horizontal grid is $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

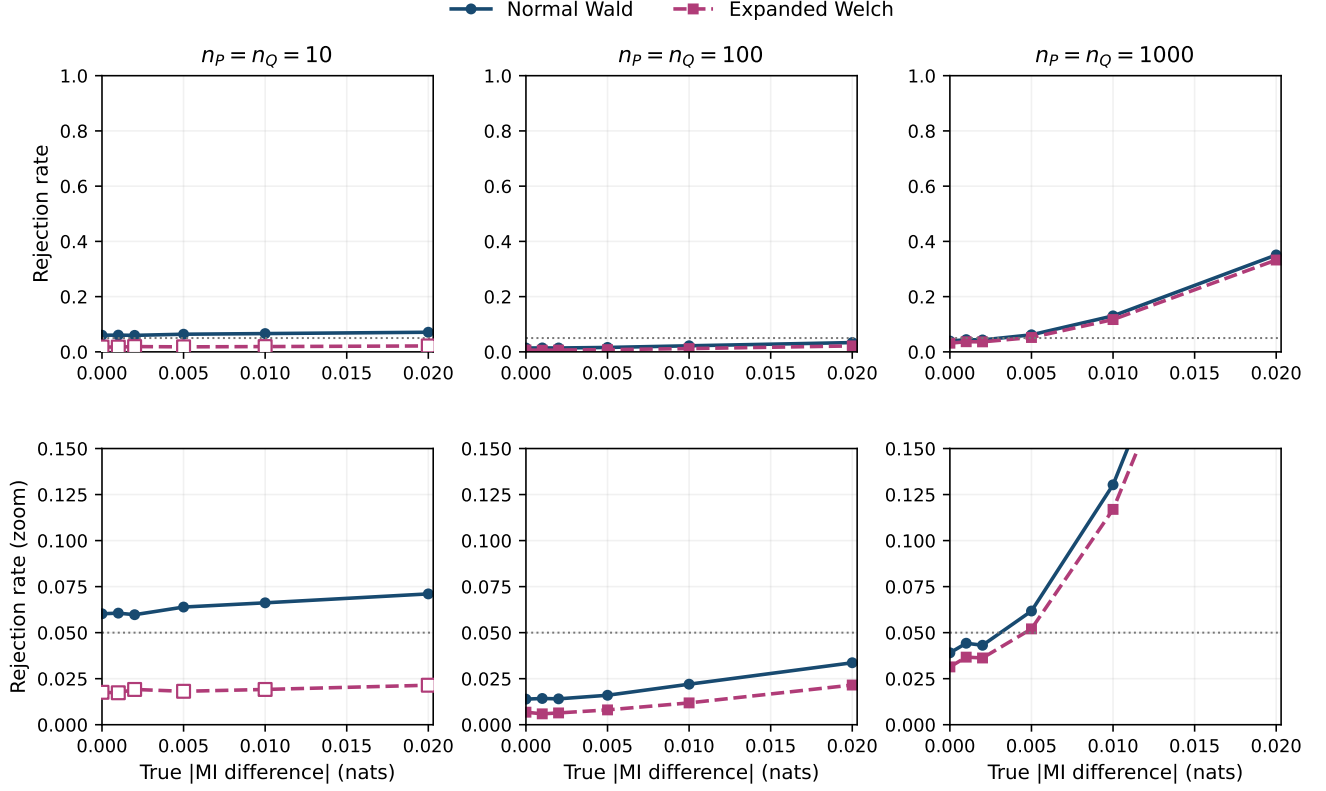


Figure 6.5: 3×3 tables, both P and Q have row and column margins $m_3(0.8) = (0.8, 0.1, 0.1)$. Columns: $n_P = n_Q = 10, 100, 1000$, respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3); $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The horizontal grid is $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

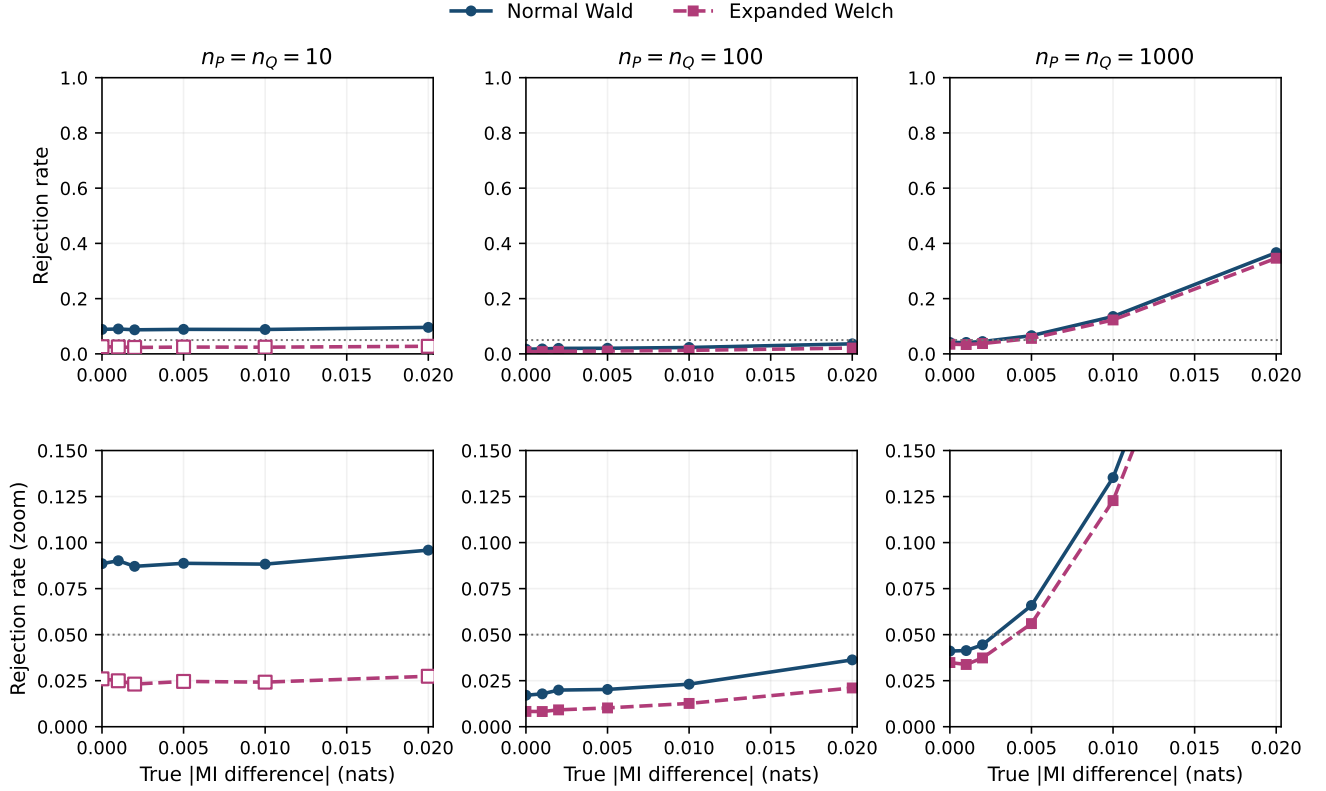


Figure 6.6: 3×3 tables, P has row and column margins $m_3(0.7) = (0.7, 0.15, 0.15)$, while Q has $m_3(0.8) = (0.8, 0.1, 0.1)$. Columns: $n_P = n_Q = 10, 100, 1000$, respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3); $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The horizontal grid is $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

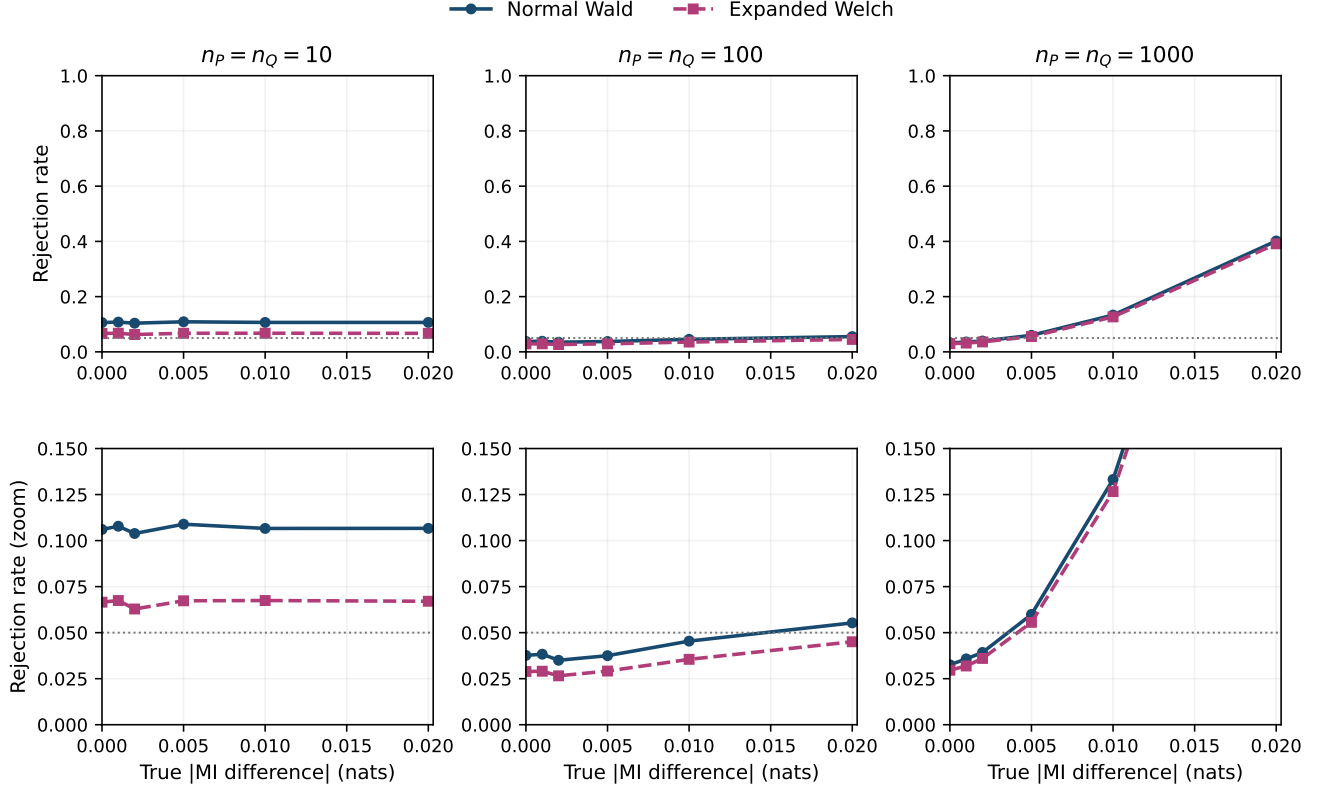


Figure 6.7: 5×5 tables, both P and Q have uniform row and column margins $(1/5, \dots, 1/5)$. Columns: $n_P = n_Q = 10, 100, 1000$, respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3); $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The horizontal grid is $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

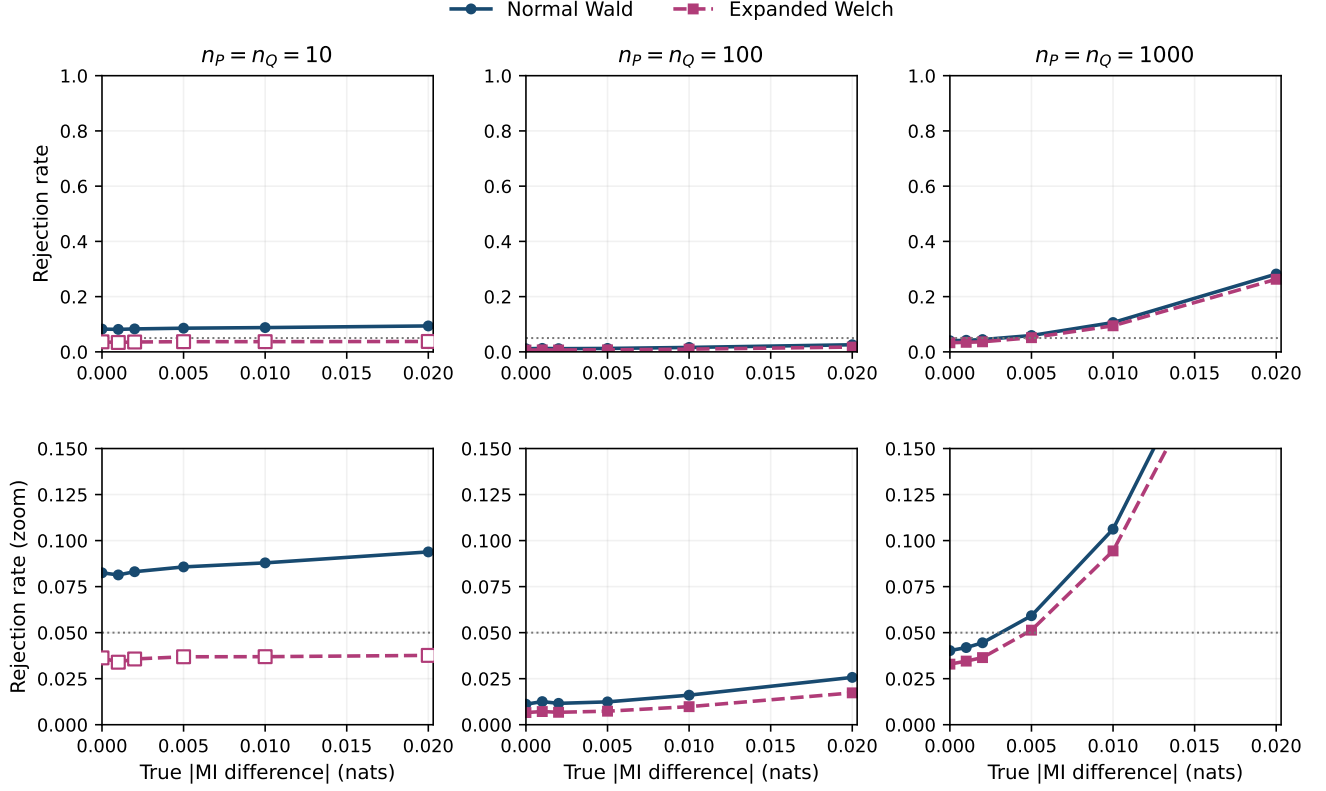


Figure 6.8: 5×5 tables, both P and Q have row and column margins $m_5(0.8) = (0.8, 0.05, 0.05, 0.05, 0.05)$. Columns: $n_P = n_Q = 10, 100, 1000$, respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3); $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The horizontal grid is $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

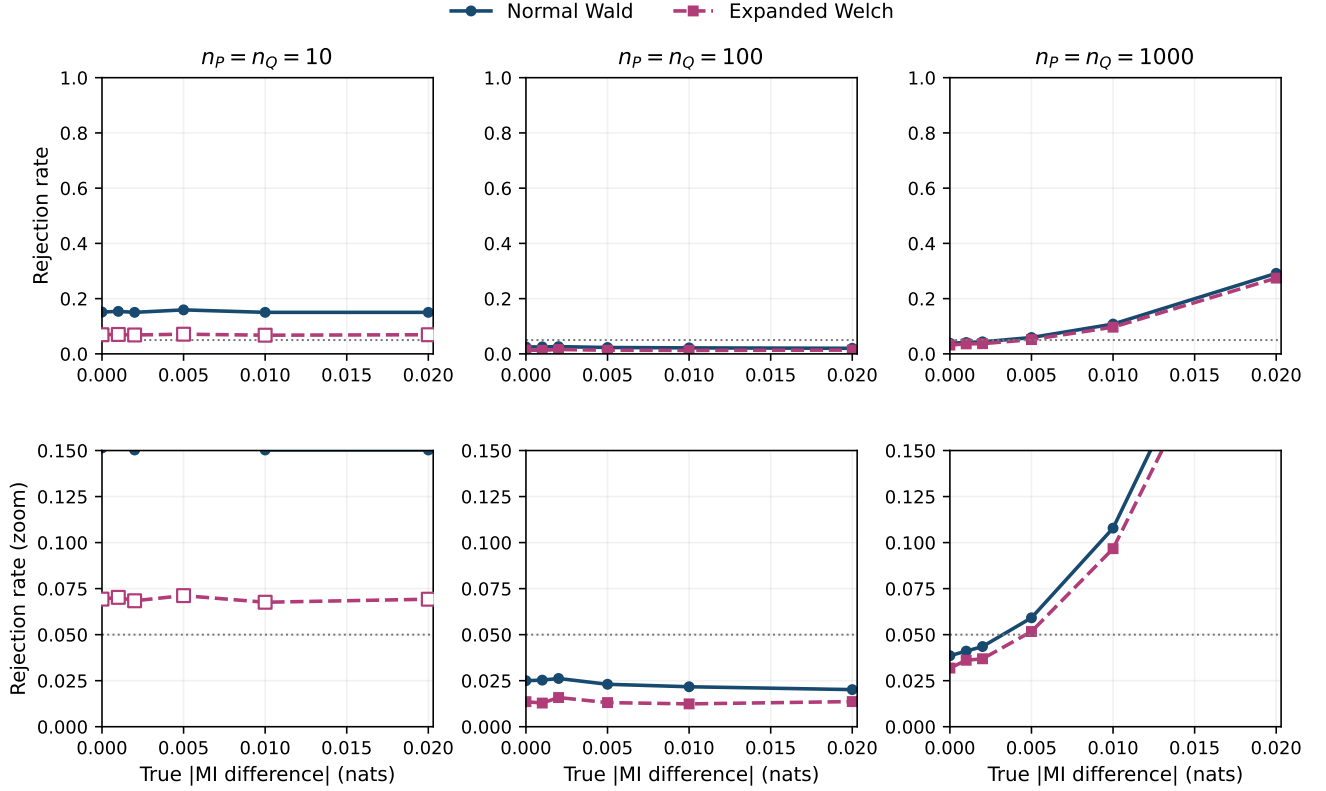


Figure 6.9: 5×5 tables, P has row and column margins $m_5(0.7) = (0.7, 0.075, 0.075, 0.075, 0.075)$, while Q has $m_5(0.8) = (0.8, 0.05, 0.05, 0.05, 0.05)$. Columns: $n_P = n_Q = 10, 100, 1000$, respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3); $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The horizontal grid is $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

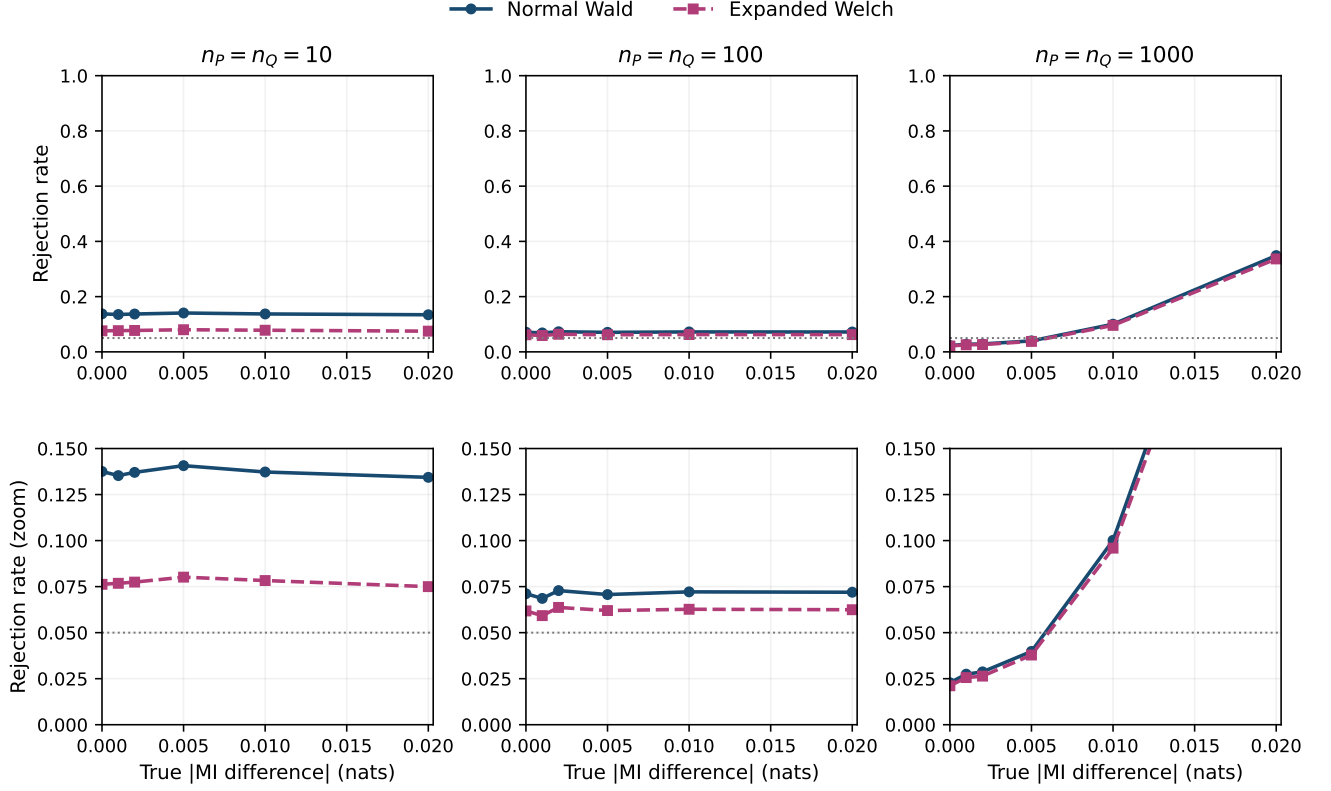


Figure 6.10: 8×8 tables, both P and Q have uniform row and column margins $(1/8, \dots, 1/8)$. Columns: $n_P = n_Q = 10, 100, 1000$, respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3); $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The horizontal grid is $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

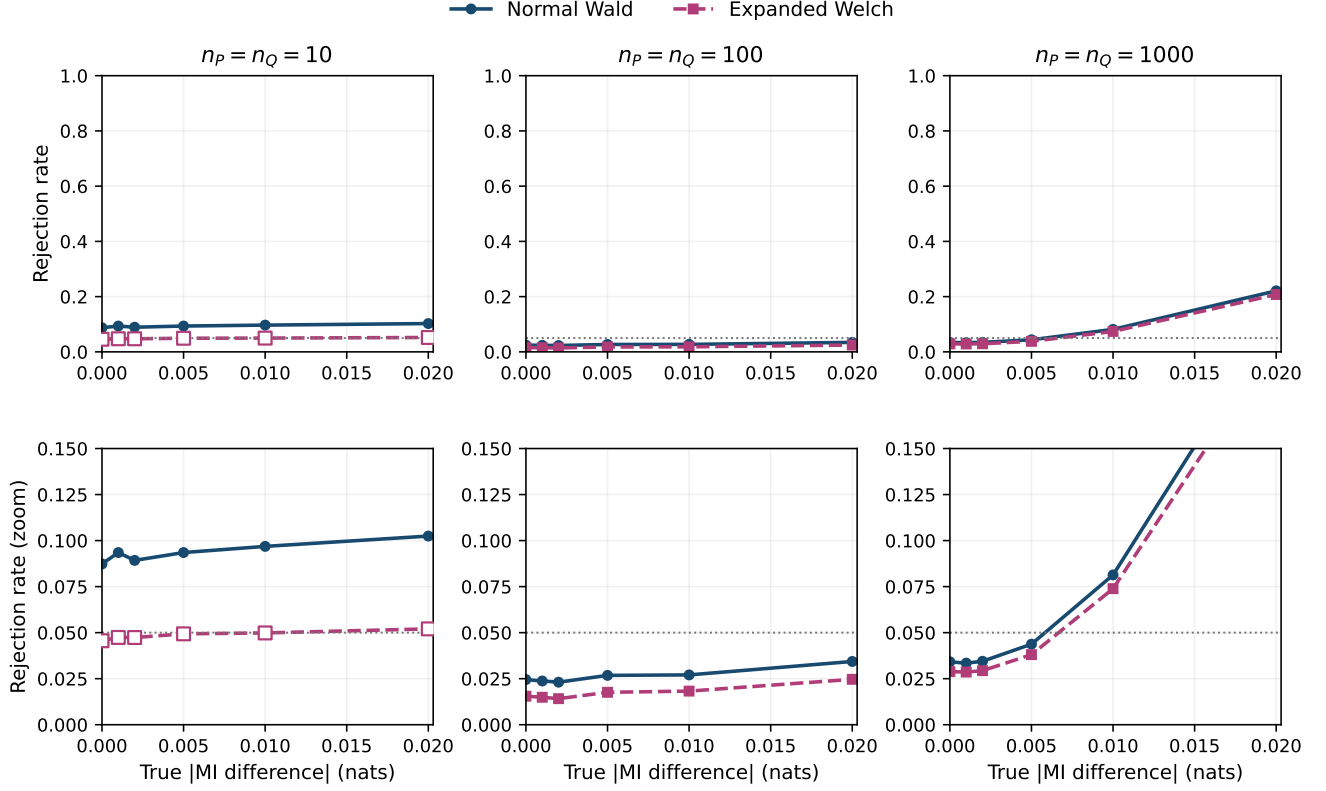


Figure 6.11: 8×8 tables, both P and Q have row and column margins $m_8(0.8) = (0.8, 0.2/7, \dots, 0.2/7)$. Columns: $n_P = n_Q = 10, 100, 1000$, respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3); $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The horizontal grid is $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

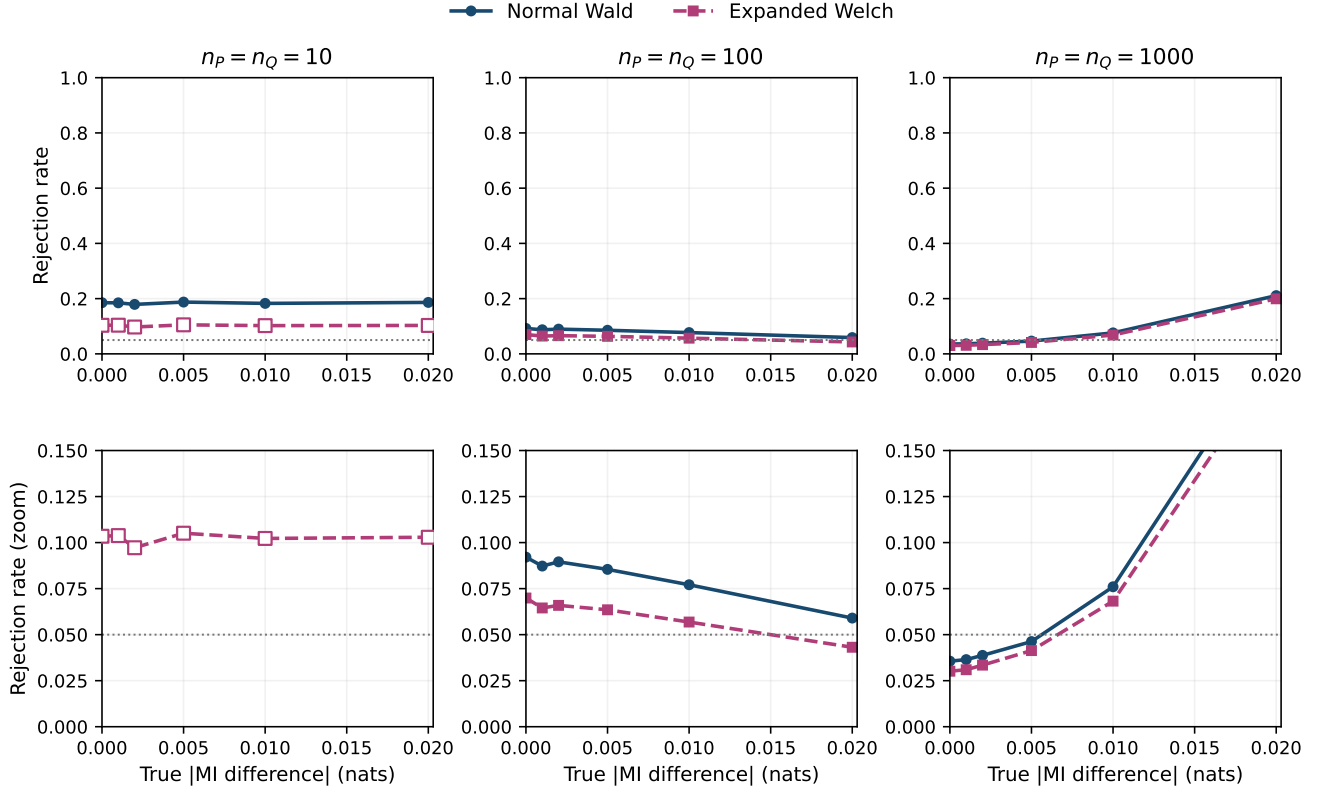


Figure 6.12: 8×8 tables, P has row and column margins $m_s(0.7) = (0.7, 0.3/7, \dots, 0.3/7)$, while Q has $m_s(0.8) = (0.8, 0.2/7, \dots, 0.2/7)$. Columns: $n_P = n_Q = 10, 100, 1000$, respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3); $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The horizontal grid is $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

with $n = 10$ or 20 , Expanded Welch is numerically closer to 0.05 in 15 and Wald in 9 . This is a descriptive count, not a test of independent improvements. The trend reverses in the 24 main null settings at $n = 500$ or $1,000$: Wald is closer to 0.05 in all 24 . At $n = 1,000$, Wald null rates across the main settings lie between 0.0228 and 0.0453 ; Expanded Welch lies between 0.0211 and 0.0393 . Neither is uniformly at the nominal level.

The interpretation is directional. Expanded Welch uses a heavier-tailed reference and therefore lowers rejection; it does not diagnose whether Wald started above or below 0.05 . The same adjustment can consequently improve a liberal regime and worsen a conservative one. Closeness to 0.05 must be read regime by regime and not treated as an inherent advantage of either method.

Two fully valid null points make the contrast concrete. With uniform 8×8 populations, $n_P = n_Q = 20$, and equal true MI of 0.02 , Wald rejects 0.06870 of samples and Expanded Welch 0.05695 . The latter is closer to 0.05 . With uniform 2×2 populations at $n_P = n_Q = 1,000$, the respective rates are 0.04450 and 0.03865 ; the correction moves an already conservative procedure farther below the target. Both methods are valid on all replicates in these examples, so the change is attributable to their different references, not missing outputs.

These two examples isolate the central trade-off. In the 8×8 case, the lower rejection rate corrects part of Wald's excess. In the 2×2 case, the same downward movement is harmful because Wald was already conservative. Expanded Welch is therefore a one-direction adjustment, not an adaptive calibration procedure.

Table 6.1: Two fully valid main-grid null regimes. Brackets beside each rate are pointwise Wilson 95% intervals; the final bracket is the approximate paired 95% interval for Wald minus Expanded Welch on the same samples. Source configurations are `config_313a5e5c05280f9e` and `config_700945d218841f2e`, respectively.

Regime	Wald	Expanded	Wald minus Expanded
	Rate [95% interval]	Rate [95% interval]	Rate [95% interval]
8×8 , uniform, $n = 20$	0.0687 [0.0653, 0.0723]	0.0570 [0.0538, 0.0602]	0.0118 [0.0103, 0.0132]
2×2 , uniform, $n = 1000$	0.0445 [0.0417, 0.0474]	0.0387 [0.0361, 0.0414]	0.0059 [0.0048, 0.0069]

The lower sample-size figures also show an important limit to interpreting small rejection rates. In the 8×8 different-skew alternative with $n_P = n_Q = 5$, $I(P) = 0.02$ and $I(Q) = 0.021$, Wald rejects 0.30865 and Expanded Welch rejects 0.00000 , but their valid-result rates are only 0.83840 and 0.31415 . These are probabilities of a

valid rejection, not a clean comparison of power at matched achieved false-positive rates. The corresponding null and validity records must be examined before calling the gap a power advantage.

6.2 Baseline MI and the location of changed cells

The main grid fixes $I(P) = 0.02$. Does starting closer to independence change the conclusion? Figure 6.13 holds shape, margins, sample size and changed cells fixed while varying only the common starting MI. The zero-difference points are especially informative: for $I_0 = 0.0001, 0.001, 0.02$, Wald's rates are 0.0193, 0.0146 and 0.0171, while Expanded Welch's are 0.0017, 0.0016 and 0.0083. All the plotted null calculations in this focused set are valid. Replacing the normal reference with a heavier-tailed Student reference does not address the already-low null rejection in these settings.

Starting closer to independence therefore does not create a general case for Expanded Welch. Here the main difficulty is already present in the first-order statistic and standard error; using a larger Student critical value only makes the conservative rejection rate smaller.

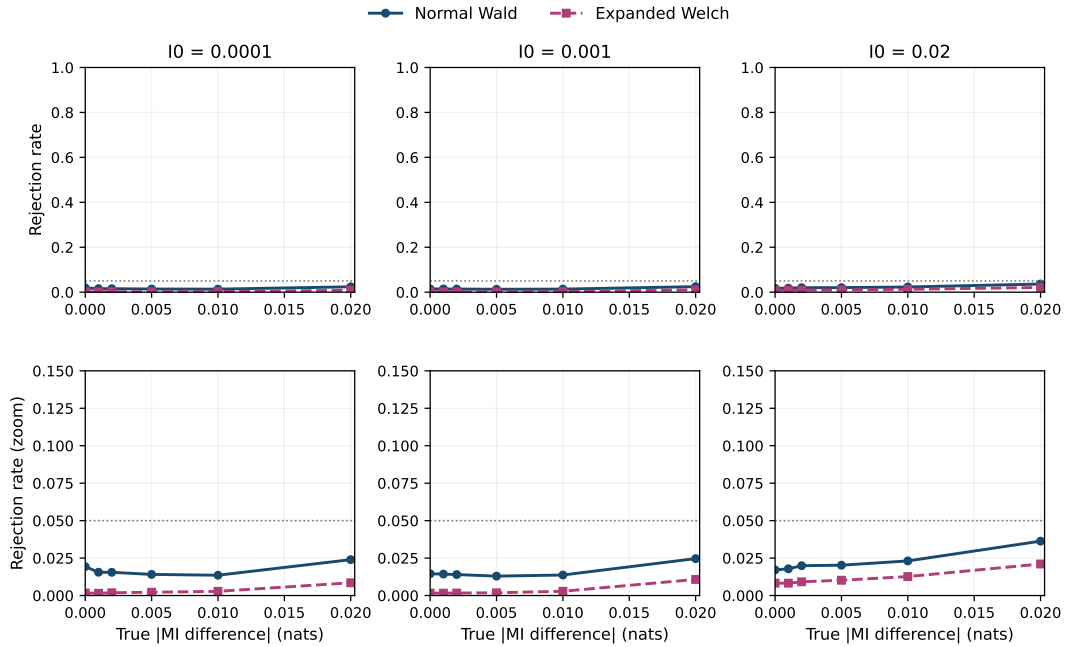


Figure 6.13: Baseline-MI comparison for 3×3 tables. Columns have $I(P) = 0.0001, 0.001, 0.02$ nats, respectively; $I(Q) = I(P) + |\Delta|$. P has row and column margins $(0.7, 0.15, 0.15)$ and Q has $(0.8, 0.1, 0.1)$. Both use the first-cell change block, $n_P = n_Q = 100$, and $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. Top: 0–1 rejection scale; bottom: 0–0.15 zoom. Each point has 20,000 replicates; hollow markers flag validity below 90%.

Figure 6.14 asks whether the association being placed in an early, rare or distributed set of cells changes performance when the margins and target MI are fixed. At the displayed null points, Wald rates are 0.0169, 0.0170 and 0.0160 and Expanded Welch rates are 0.0039, 0.0044 and 0.0041 for first, rare and spread patterns. The similarity of those three null rates in this selected setting is not evidence that the patterns are equivalent in all tables. The full 3×3 , 5×5 and 8×8 matched sets are in the atlas.

Within this controlled comparison, moving the association to different cells does not remove the conservative gap between the methods. The implication is limited but useful: the displayed result is not an artefact of only the first 2×2 change block used in the main landscape.

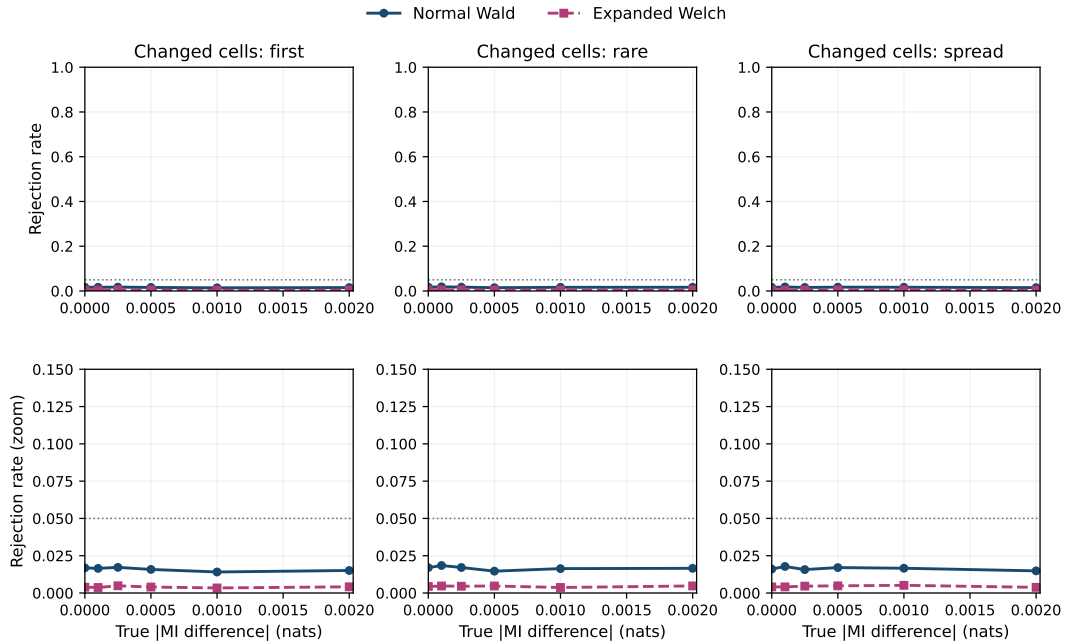


Figure 6.14: Changed-cell comparison for 5×5 tables. Columns place association in the first 2×2 block, last (rare) 2×2 block, or a centred score outer product spread across all cells. For every column, P 's row and column margins are $(0.7, 0.075, 0.075, 0.075, 0.075)$, and Q 's are $(0.8, 0.05, 0.05, 0.05, 0.05)$. Here $I(P) = 0.0001$, $I(Q) = I(P) + |\Delta|$, $n_P = n_Q = 100$, and $|\Delta| = 0, 0.0001, 0.00025, 0.0005, 0.001, 0.002$ nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates.

6.3 Unequal samples and the direction of the difference

Figures 6.15 and 6.16 hold the unordered sample sizes at 50 and 250 but reverse which population has the larger sample. They also show the two directions of the MI

difference separately. The distinction matters because the finite bias corrections are $d/(2n_P)$ and $d/(2n_Q)$, and the two populations have different margins. For example, at $(n_P, n_Q) = (50, 250)$ and a 0.02-nat difference, Wald's rejection is 0.0994 when Q has higher MI and 0.0495 when P has higher MI. Expanded Welch gives 0.0247 and 0.0232. All four rates in that specific comparison have validity one. These alternative rates should still be read beside the matched null points; they are not equal-size power.

The practical implication is that an absolute MI difference does not define a complete testing problem. Which population has the higher MI and which one has the larger sample changes the bias and variance contributions. Power should therefore not be averaged across the two directions of sample imbalance.

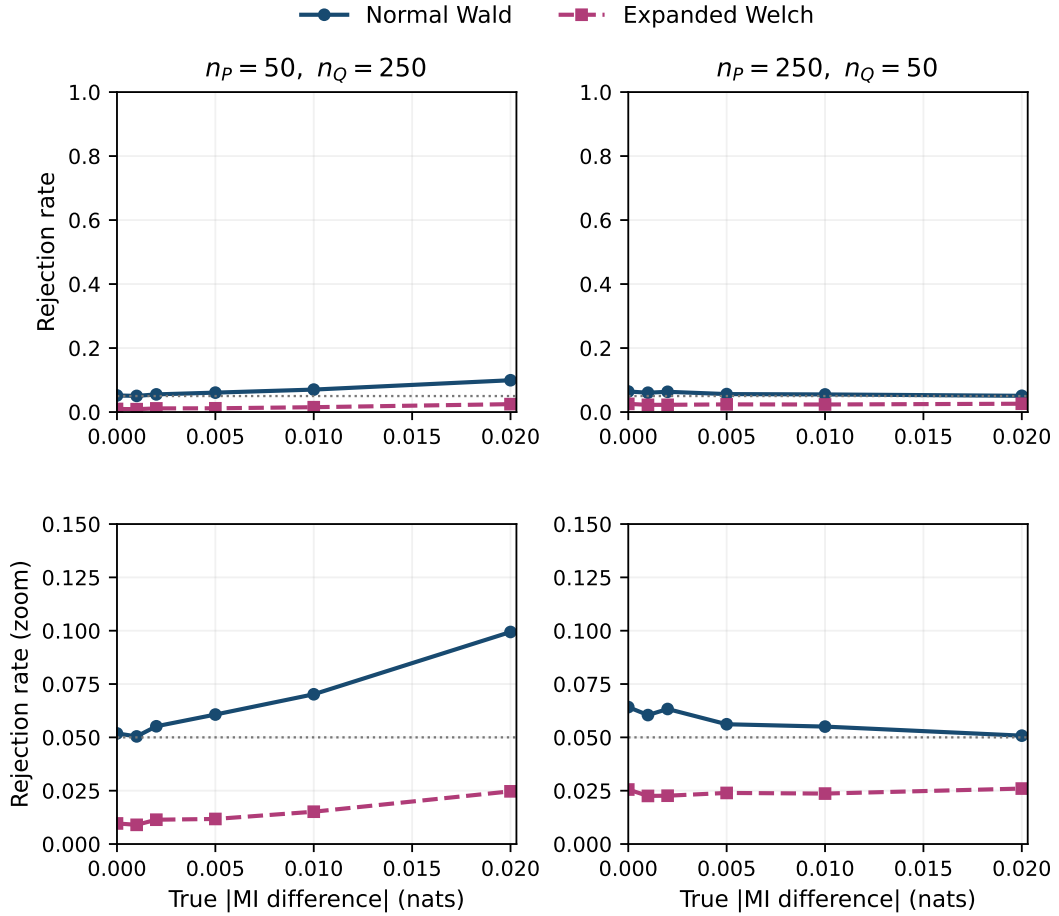


Figure 6.15: Unequal samples, Q higher. Columns have $(n_P, n_Q) = (50, 250)$ and $(250, 50)$. For both columns, P has 3×3 row and column margins $(0.7, 0.15, 0.15)$ and Q has $(0.8, 0.1, 0.1)$; the first-cell change block is used. $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$, with $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates.

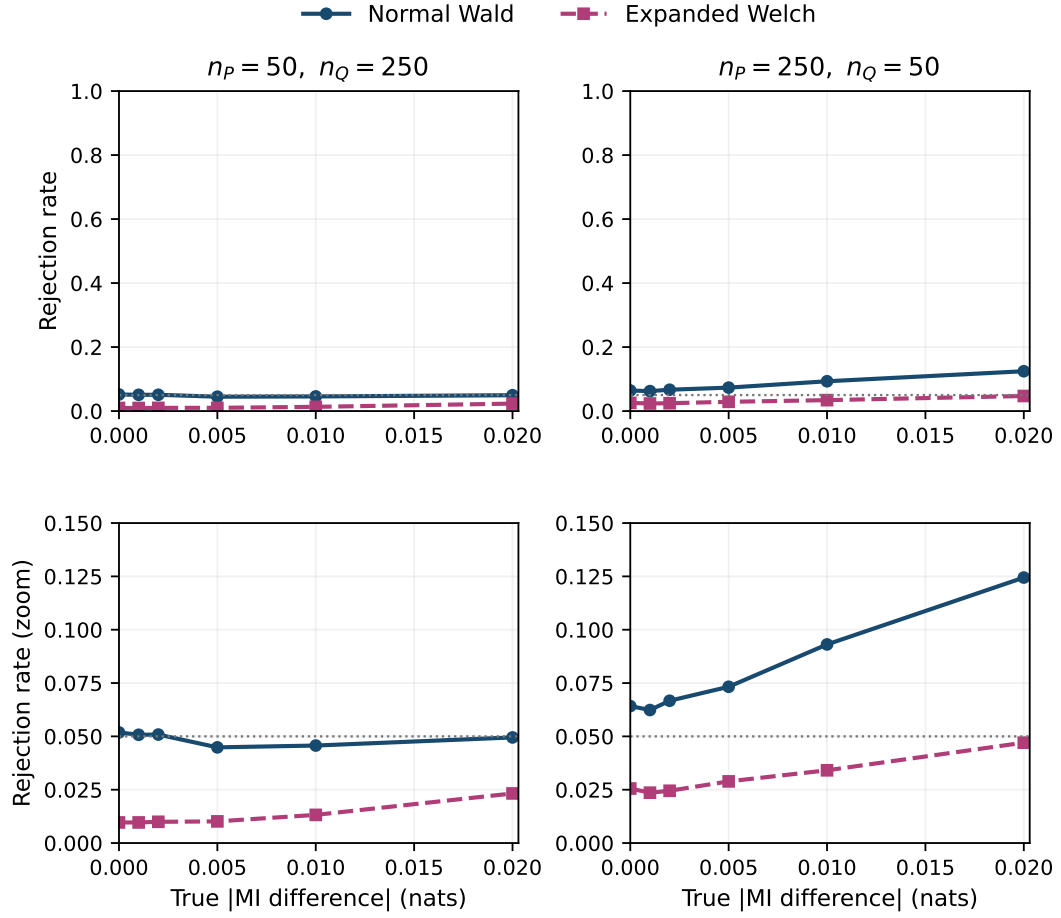


Figure 6.16: Unequal samples, P higher. Columns have $(n_P, n_Q) = (50, 250)$ and $(250, 50)$. P 's 3×3 row and column margins are $(0.7, 0.15, 0.15)$; Q 's are $(0.8, 0.1, 0.1)$. Both use the first-cell change block. $I(Q) = 0.02$ and $I(P) = 0.02 + |\Delta|$, with $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates.

6.4 Larger differences and rectangular tables

The main grid deliberately concentrates on small MI differences. A separate 2×2 family extends the feasible axis to 0.2 nats without changing the axis units or pretending unconstructed targets exist. This shows where detection eventually rises, while retaining the zero-difference point that defines null calibration. Rectangular 2×3 and 3×5 tables check that the discussion is not tied only to square alphabets. The panels do not average over shapes.

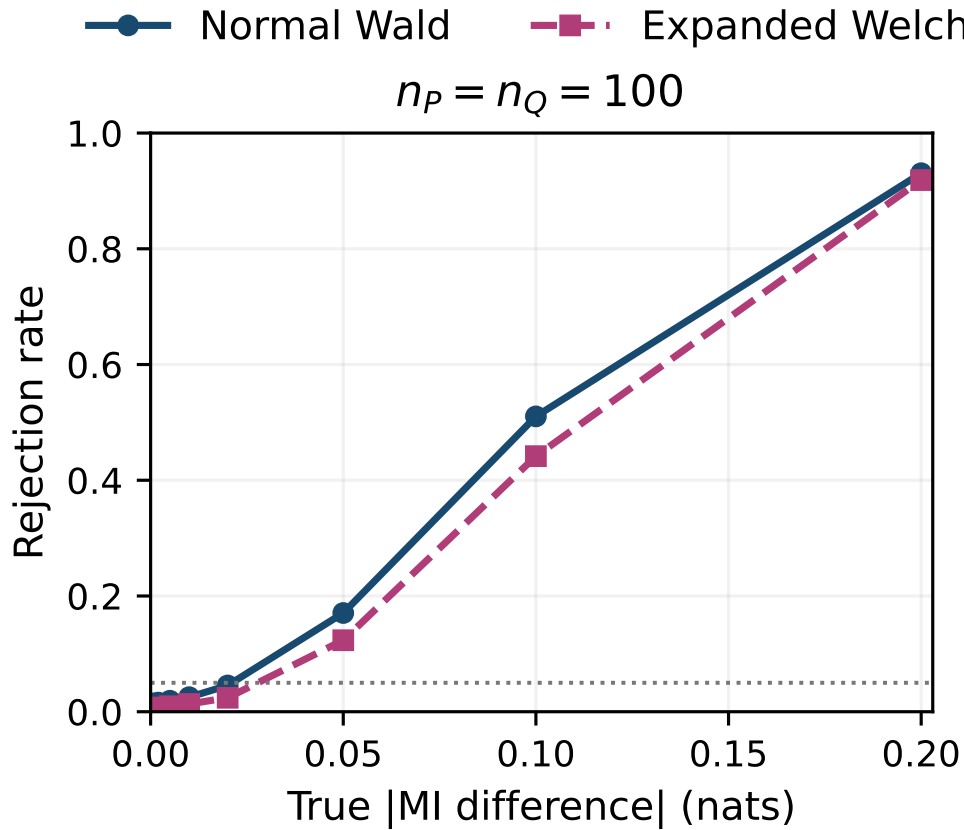


Figure 6.17: Wider MI differences in a uniform 2×2 population pair. The single panel has P and Q row and column margins $(0.5, 0.5)$, first-cell changes, $n_P = n_Q = 100$, $I(P) = 0.02$, and $I(Q) = 0.02 + |\Delta|$. The horizontal grid is $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2$ nats; the rejection axis is 0–1. Each point has 20,000 replicates.

The wider 2×2 curve shows that both methods eventually detect large MI differences, while Expanded Welch remains below Wald until the rejection rates approach one. This is the expected cost of its larger critical value: the power loss matters most at intermediate differences and becomes small when the signal is overwhelming.

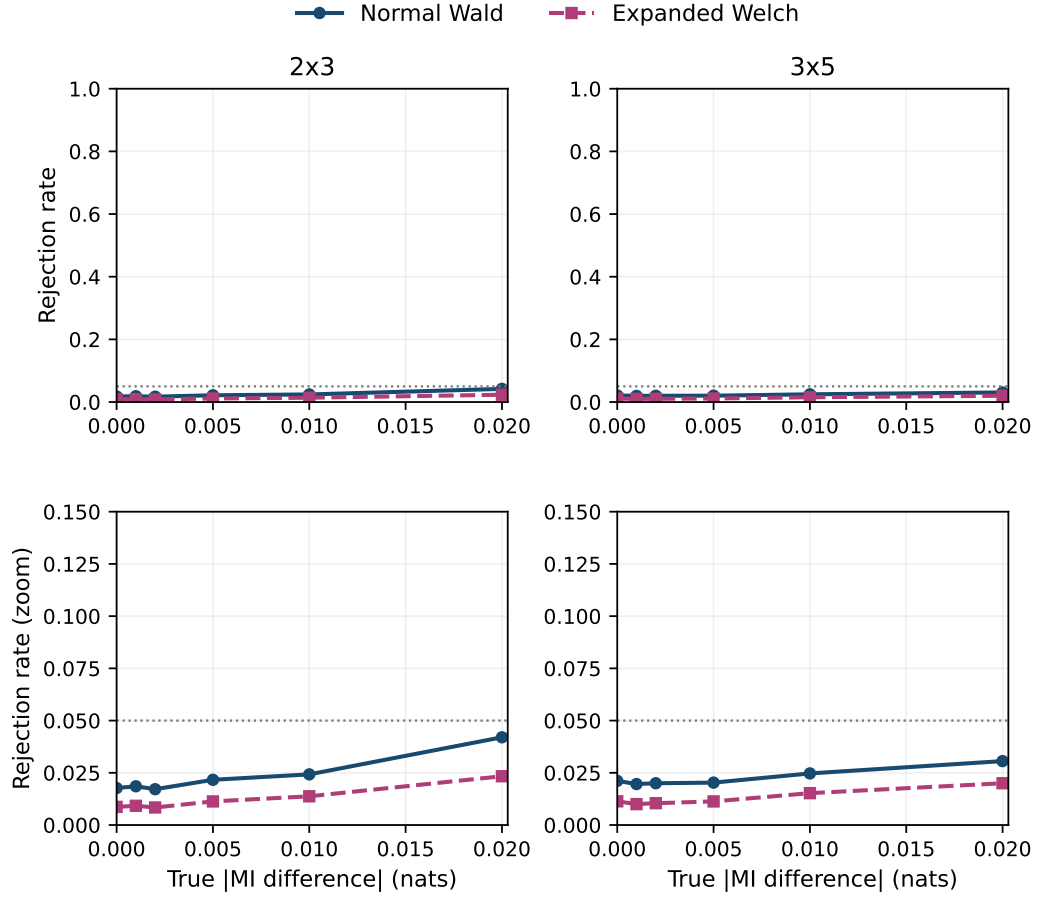


Figure 6.18: Rectangular tables. Columns are 2×3 and 3×5 . In each dimension k , P 's corresponding margin is $m_k(0.7)$ and Q 's is $m_k(0.8)$, for both rows and columns. Thus the first column uses P rows $(0.7, 0.3)$, columns $(0.7, 0.15, 0.15)$, and Q rows $(0.8, 0.2)$, columns $(0.8, 0.1, 0.1)$; the second applies the same formula to 3 and 5 categories. Both use first-cell changes, $n_P = n_Q = 100$, $I(P) = 0.02$, and $I(Q) = 0.02 + |\Delta|$ for $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates.

In the rectangular tables, both methods remain below 0.05 throughout the displayed range, with Expanded Welch lower still. Those low alternative rejection rates indicate weak detection, not good null calibration. Together the two checks show that the main trade-off is not confined to square tables or to the narrow MI grid.

6.5 Population construction sensitivity

The additive block is interpretable, but it is only one way to make tables with fixed margins and target MI. Figure 6.19 compares it with an ordinal log-linear construction under matched targets. The null rates are not identical: for the displayed 5×5 different-skew comparison at $n = 100$, Wald gives 0.0250 under additive construction and 0.0181 under log-linear construction; Expanded Welch gives 0.0136 and 0.0119. All are below nominal and all are valid. Thus matching the scalar MI and margins does not fix the complete finite-sample behaviour. The constructor comparison is a sensitivity check, not proof that the population families exhaust all relevant dependence structures.

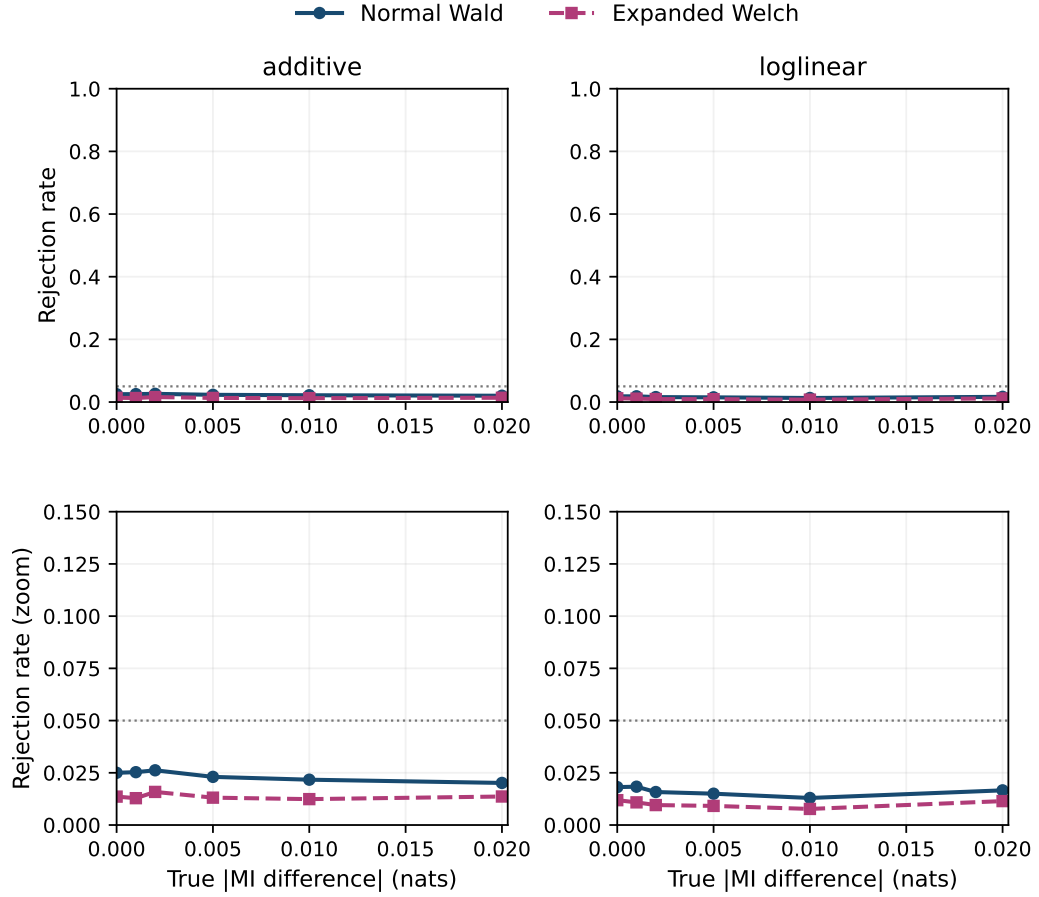


Figure 6.19: Additive versus ordinal log-linear construction for 5×5 tables. The left column uses the first-cell additive block; the right fits an ordinal interaction while preserving the margins. In both, P 's row and column margins are $(0.7, 0.075, 0.075, 0.075, 0.075)$ and Q 's are $(0.8, 0.05, 0.05, 0.05, 0.05)$. Both have $I(P) = 0.02$, $I(Q) = 0.02 + |\Delta|$, $n_P = n_Q = 100$, and $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates.

6.6 Extreme skew, independence and larger samples

The extreme-skew check deliberately keeps configurations with very small expected cell counts. At $n_P = n_Q = 100$, 3×3 tables with dominant margins 0.999 for P and 0.9995 for Q yield valid null results in only 0.0118 of Wald replicates and 0.00005 of Expanded Welch replicates. Both unconditional false-positive rates are zero, but this is mostly an absence of valid tests, not evidence of excellent calibration. With dominant margins 0.9 and 0.95, the corresponding null rates are 0.0701 and 0.0088; Expanded Welch lowers an excessive Wald rate past the target. Figure 6.20 shows all three specified skew levels, not just the favourable one.

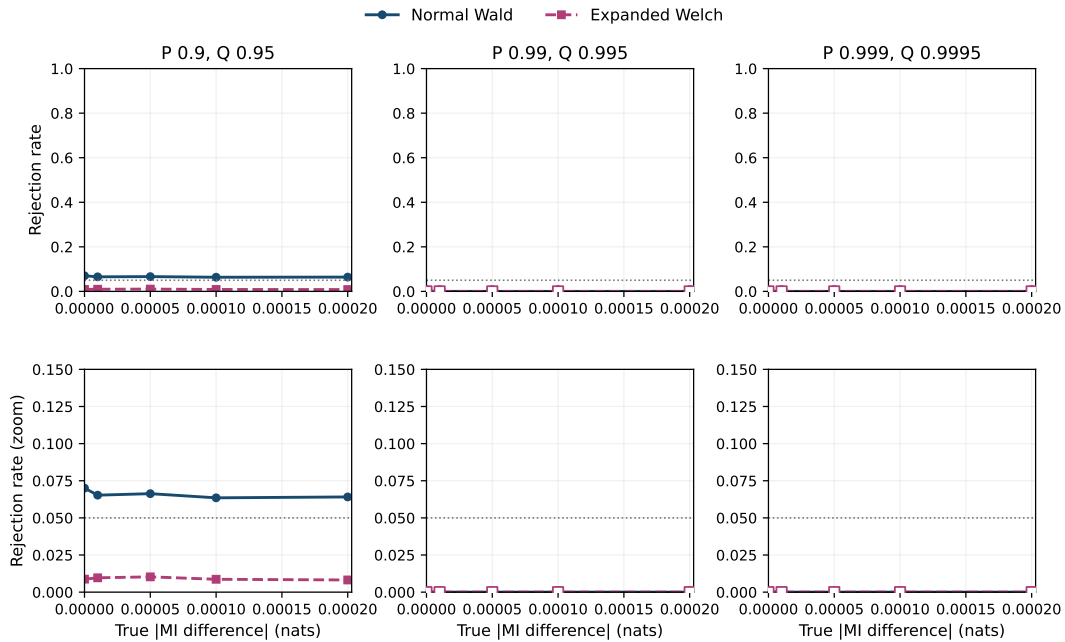


Figure 6.20: Extreme-skew 3×3 tables. Across columns the dominant row and column probabilities for (P, Q) are $(0.9, 0.95)$, $(0.99, 0.995)$, and $(0.999, 0.9995)$. Each remaining margin entry splits the leftover probability equally between the two other categories. Both use first-cell changes, $n_P = n_Q = 100$, $I(P) = 0.00001$, and $I(Q) = I(P) + |\Delta|$, for $|\Delta| = 0, 0.00001, 0.00005, 0.0001, 0.0002$ nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates; hollow markers flag validity below 90%.

Exact independence is a separate theoretical boundary. Figure 6.21 starts P at MI zero and includes the equal-MI point where both populations are independent. This is not a routine positive-MI validation of Expanded Welch: its population first-order variance is zero at independence. At $n = 100$, for example, the uniform 2×2 null rates are only 0.00025 and 0.00010 for Wald and Expanded Welch. Appendix D explains why a second-order independence test has a different reference.

Does increasing n resolve the null error when the populations remain fixed?

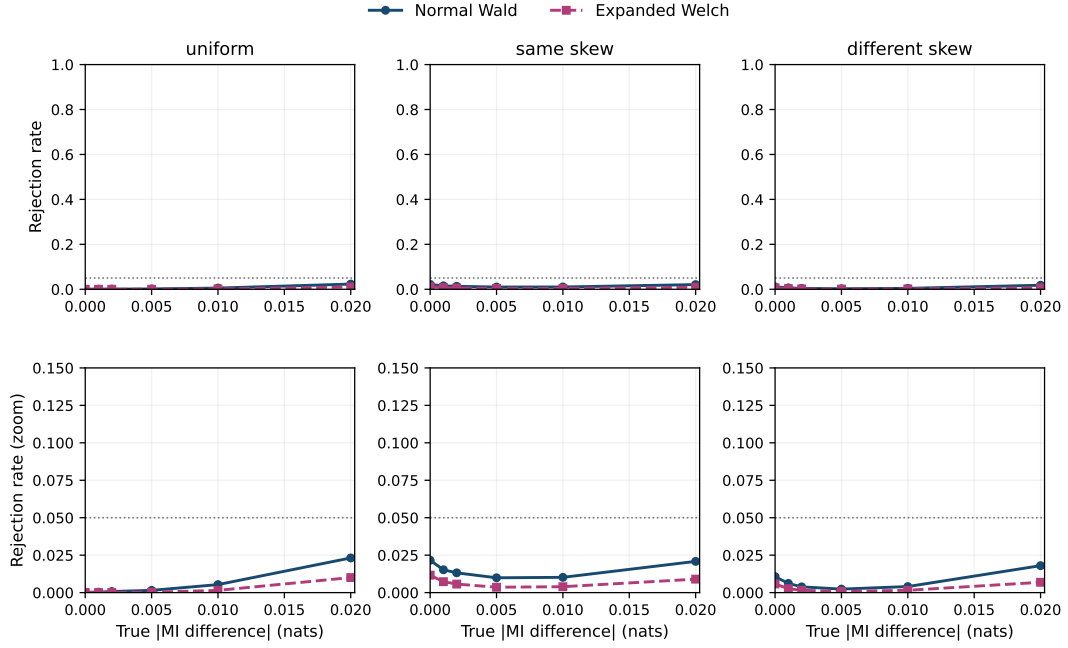


Figure 6.21: Exact-independence boundary for 2×2 tables. Columns are uniform margins $(0.5, 0.5)$ for both populations, same skew $(0.8, 0.2)$ for both, and different skew with $P = (0.7, 0.3)$ and $Q = (0.8, 0.2)$; each vector is used for both rows and columns. The first-cell block is used, $I(P) = 0$, $I(Q) = |\Delta|$, $n_P = n_Q = 100$, and $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates.

Figure 6.22 keeps equal MI and changes only sample size. Across the 12 first-block profiles with $I(P) = I(Q) = 0.02$ and $n = 50,000$, Wald’s false-positive rates range from 0.0470 to 0.0527 and Expanded Welch’s from 0.0466 to 0.0526. At equal MI 0.0001, however, the ranges at the same n are 0.0089–0.0280 and 0.0076–0.0138. These are finite-sample observations for fixed populations. They show that very small positive MI can remain difficult at a large sample size, not that the regular asymptotic theory is mathematically false.

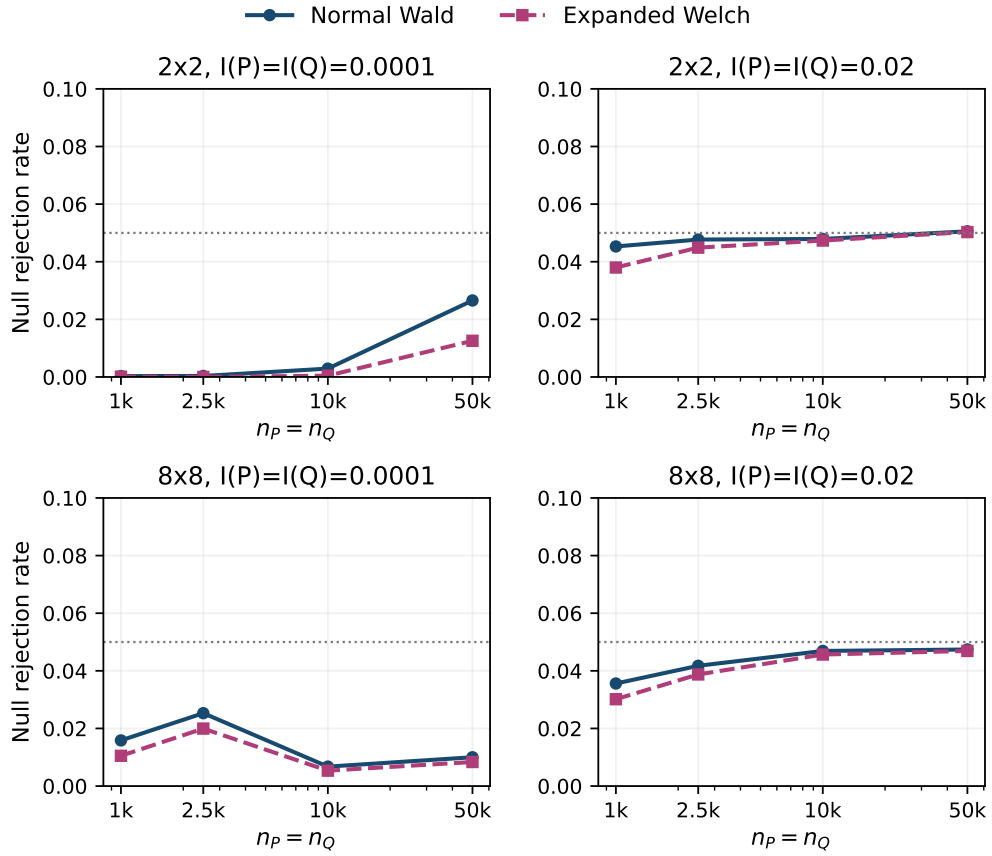


Figure 6.22: Large-sample null behaviour. Rows compare 2×2 and 8×8 tables; columns have common true MI 0.0001 and 0.02 nats. For both shapes, P 's row and column margins are $m_k(0.7)$ and Q 's are $m_k(0.8)$, with $k = 2$ or 8 . Both populations use the first-cell block. The horizontal axis is the equal sample size $n_P = n_Q = 1,000, 2,500, 10,000, 50,000$ on a labelled log scale; the vertical axis is null rejection from 0 to 0.10. Each point has 20,000 replicates.

6.7 Overall calibration and a mechanism check

The main-grid null contains two different hypotheses in terms of complete population equality. The 72 uniform and same-skew configurations have $P = Q$, whereas the 36 different-skew configurations have $P \neq Q$ but $I(P) = I(Q)$. The latter are the weak-null cases that most directly match the thesis motivation. Table 6.2 reports these profiles separately. The broad interval 0.025–0.075 is used only to describe the landscape; it is not a formal acceptance region for calibration.

Table 6.2: Location, separately for each method, of the 108 main-grid null configurations relative to the descriptive rejection-rate interval 0.025–0.075. Each profile contains 36 configurations. The final column separately counts configurations with validity below 90%; it overlaps the three rate-location columns. In particular, all four $n = 2$ configurations per profile have zero validity for both methods. Uniform and same-skew are strong nulls; different-skew is the equal-MI weak null. Exact rates and validity are available in the saved results described in Appendix C.

Margins	Method	Below	Inside	Above	Validity below 90%
Uniform	Normal Wald	10	18	8	4
Uniform	Expanded Welch	19	16	1	5
Same skew	Normal Wald	14	15	7	8
Same skew	Expanded Welch	25	11	0	12
Different skew	Normal Wald	9	16	11	8
Different skew	Expanded Welch	19	13	4	12

The table makes the direction of the adjustment clear. Expanded Welch greatly reduces the number of rates above the descriptive band, but it also increases the number below it in every margin profile. Its apparent protection against liberal rejection is therefore purchased with additional conservatism rather than with calibration in both directions.

A separate supplementary study tested the mechanism without altering the confirmatory results. For the representative 3×3 different-skew null with $I(P) = I(Q) = 0.02$ and $n = 100$, Wald rejection was 0.0166. Using an independent Monte Carlo estimate of the numerator’s finite-sample standard deviation gave 0.0551, while using the population first-order standard deviation gave 0.0976. The first-order approximation is therefore imperfect in both the estimated standard error and the numerator distribution. This supports the simpler interpretation that changing only the reference distribution cannot correct every finite-sample error. Full pilot and ablation results are retained in the experiment supplement.

6.8 Runtime and summary of evidence

The complete scalar calls have the same $O(rc)$ asymptotic complexity. The observed median time across the 60 timing regimes is about 85 microseconds for Wald and 150 microseconds for Expanded Welch. The median Expanded-to-Wald ratio is about 1.75, with regime ratios about 1.72–1.79. For the explicitly timed 3×3 , different-skew, $n_P = n_Q = 100$, equal-MI input regime, the medians are 85.083 and 149.979 microseconds. These are implementation- and machine-specific timings, not universal complexity constants. Exact per-regime medians and interquartile ranges are in the saved runtime file cited in Appendix C.

Computationally, the correction is feasible: it has the same cell-linear order as Wald and adds roughly 75% to a scalar call in this implementation. The overhead is modest compared with resampling, but it is still an extra cost. Because the statistical results do not show a general improvement, runtime does not provide a separate reason to prefer Expanded Welch.

Table 6.3: Complete scalar-call times in microseconds for the different-skew null at $n_P = n_Q = 100$ and $I(P) = I(Q) = 0.02$. Entries give median [interquartile range] over five passes on 200 saved inputs per regime. Population construction and multinomial sampling are excluded.

Shape	Normal Wald	Expanded Welch	Expanded/Wald
2×2	85.8 [85.5, 86.3]	150.0 [149.4, 151.3]	1.75
3×3	85.1 [84.6, 85.6]	150.0 [148.9, 150.9]	1.76
3×5	85.2 [84.9, 85.5]	150.1 [149.5, 150.9]	1.76
5×5	85.3 [85.1, 85.7]	149.2 [148.5, 150.1]	1.75
8×8	88.0 [87.1, 88.9]	153.4 [152.4, 154.8]	1.74

The evidence separates three observations. First, Expanded Welch can make a liberal Wald null rate closer to 0.05. Second, because it uses the same statistic with a heavier-tailed reference, it can also reduce rejection where Wald is already conservative. Third, in sparse samples the unconditional rate can fall because the expanded calculation is invalid more often. No one number summarises these behaviours; the matched curves and their validity records are the basis for the discussion that follows.

Chapter 7

Discussion

The proposed correction has a clear mathematical purpose: it represents sampling uncertainty in the MI variance estimate before using that estimate in a two-sample test. The experiments show that the purpose is narrower than its possible practical effect. A better description of the denominator does not by itself repair every finite-sample error in the numerator, in the first-order approximation, or in the availability of a valid statistic.

7.1 Answers to the research questions

The first research question was whether an MI-specific effective degree of freedom can be derived. Chapter 4 does so by differentiating the variance of pointwise MI along one-cell probability perturbations. The derivative includes changes to row and column margins, not just to the selected cell. Its sampling variance supplies the component degrees of freedom used in Satterthwaite moment matching. These derivative and moment-matching steps define a reproducible statistic. The final Student reference remains a working approximation, not an exact finite-sample theorem for MI.

The second question concerned false positives, detection and validity. Figure 6.10 contains a fully valid small-sample null case where Expanded Welch moves an excessive Wald false-positive rate toward 0.05. Figure 6.1 contains a larger-sample null case where the same adjustment moves a below-nominal Wald rate farther from 0.05. In the sparsest panels, a low unconditional rejection rate may largely reflect invalid outputs. Thus improved calibration at selected null points does not imply that the expanded procedure performs better overall. Under alternatives, its rejection rate is generally lower; these comparisons are not power at a matched achieved size.

The third question asked whether population features alter this judgement. The main landscape varies alphabet size and margins explicitly. The focused comparisons show that baseline MI, changed-cell location, sample allocation and population constructor can change the finite-sample behaviour even when the target MI difference is held fixed. The construction comparison is particularly instructive: equal margins and MI do not determine the table's complete sampling distribution. A scalar target is useful for organising experiments, but cannot guarantee identical difficulty across distinct population tables.

The fourth question concerned sample size and computation. For the tested first-block positive-MI populations with true MI 0.02, both null rates are near 0.05 at $n = 50,000$. The corresponding very-small-MI fixed populations remain noticeably conservative at that sample size. Expanded Welch has the same linear-in-cell-count order as Wald but costs about 1.75 times as much per scalar call in the measured implementation. The extra expense is modest compared with a full resampling procedure but not cost-free, particularly for many repeated comparisons.

7.2 What the correction can and cannot change

Both methods compute the same $\hat{\Delta}_{\text{BC}}$, $\widehat{\text{SE}}$ and T . With valid positive degrees of freedom, Expanded Welch has a heavier-tailed reference and a two-sided p-value no smaller than Wald's for the same table pair. This gives a direct explanation for many plots: the correction lowers rejection whether the Wald rate is too high or already too low. It is not an adaptive calibration method that can move the threshold in both directions. It also does not alter the leading bias correction or re-estimate the numerator's finite-sample distribution.

The estimated numerator and denominator come from the same count tables. Independence of the two samples allows their variance contributions to be added, but it does not make T 's numerator independent of its estimated standard error. The representative pilot in Chapter 6 shows that both parts can depart from their first-order approximations. Moment matching the denominator alone therefore cannot guarantee a Student law or correct every finite-sample error.

Near independence, the local calculation gives $V \approx 2I$, $\tau^2 \approx 4V$, and hence a population component degree of freedom near nI . This explains why the reference can become very heavy-tailed at low MI, exactly where the first-order approximation is least reliable. The result is a useful warning, not a complete finite-sample theory.

At exact independence, $I = 0$ and the population first-order MI variance vanishes. This does not mean finite empirical MI estimates are always zero. It means their

leading stochastic variation is quadratic rather than linear in the probability errors. The standard one-table likelihood-ratio statistic uses that second-order behaviour. Neither setting a reference Q to an independent distribution nor squaring the current T makes Expanded Welch the classical independence test. Appendix D gives the mathematical distinction.

7.3 Interpretation of invalid samples

The validity diagnostics matter most where expected counts are very small. Wald needs a usable total estimated variance. The expanded implementation additionally needs usable component sensitivity variances and effective degrees of freedom. An observed table can have a positive total variance while one component calculation is unusable. The final study does not silently replace these cases with a different test. Instead, it reports invalidity and counts such a replicate as a non-rejection in the primary unconditional rate.

That rate answers an operational question: how often does the complete procedure produce a valid rejection? It should not be mistaken for the conditional rejection probability among valid tables. Nor does a failure to return a p-value constitute a true negative. The saved validity and paired results allow readers to distinguish fewer rejections from fewer usable tests.

7.4 Practical conclusion and limits of the evidence

For the tested task, Normal Wald remains the simpler general baseline. Expanded Welch can be useful as a conservative sensitivity analysis in a regime where its validity is high and Wald's null rejection is demonstrably liberal. It should not be installed as a universal replacement merely because one null table shows a smaller false-positive rate. In applications, the population truth is unknown, so the most relevant calibration study is one matched to the anticipated alphabet, sample size and plausible margins. Selecting between the methods after inspecting a single p-value is not a validated selection rule.

Several limitations bound this conclusion. The population families are deliberately constructed fixed examples, not a probability sample of all joint distributions. Twenty thousand replicates make a rate at a fixed population relatively precise; they do not make the 534 selected population pairs representative of every application. The tables are at most 8×8 square and 3×5 rectangular, with the stated finite sample ladders and only a two-sided nominal level of 0.05. The main construc-

tion emphasises a local four-cell association; spread, rare-cell and log-linear checks probe but do not exhaust other possibilities. The study does not validate paired observations, continuous estimators, a real-data workflow or growing alphabets.

The protocol was frozen before the final simulation, after exploratory development of the method and design. It is therefore a reproducible final run, not an externally preregistered untouched hypothesis. Configurations displayed in several sections are the same sampled run. Repeating their figure appearances does not provide independent confirmation. The recorded Monte Carlo intervals also address repeated sampling of each fixed pair, not the broader uncertainty of generalising to an untested population.

The alternative curves use the common nominal 0.05 threshold. They show practical detection under the tested procedures, not power after giving each method a simulation-specific critical value.

Future work should address the joint finite-sample behaviour of the numerator and denominator or use a calibrated reference tailored to sparse tables. The transition near independence is a particularly distinct problem because the first-order variance tends to zero. A likelihood-ratio or resampling alternative must be assessed under the equal-MI *weak null*, not imported from a test of full distributional equality or one-table independence without checking its target. These are directions for new validation, not remedies established by the present results.

Chapter 8

Conclusion

This thesis examined whether uncertainty in an estimated MI standard error can be represented through an MI-specific Welch–Satterthwaite correction when comparing two independent categorical populations. The null is equality of their population MI values, not independence within either population or equality of the complete joint distributions.

The derivation starts from the pointwise-MI variance governing first-order sampling error. A one-cell perturbation changes the joint probabilities, the margins and the pointwise values. Differentiating the complete variance functional supplies the sampling variability needed for component effective degrees of freedom. The resulting Expanded Welch procedure retains Wald’s bias-corrected MI difference and plug-in standard error but evaluates the shared statistic against a Student reference. This is a principled moment-matching construction, not an exact finite-sample Student law.

The completed fixed-population experiment found both benefits and costs. Expanded Welch can make a liberal Wald null rate closer to the nominal 5% level. Where Wald is already conservative, the heavier-tailed reference moves rejection farther below that target. In sparse regimes, Expanded Welch also returns fewer valid results; a low unconditional rate there is not reliable evidence of good calibration. At ordinary positive MI and sufficiently large samples, both methods approached the nominal null rate in the tested fixed populations, while near-independence settings remained difficult even at large finite sample sizes. The expanded call used about 1.75 times the runtime of Wald in the measured implementation.

Near independence, the component degree of freedom is approximately nI , so Expanded Welch can impose a very heavy-tailed reference where the first-order approximation remains weak. A supplementary diagnostic confirms that errors can arise in both the estimated standard error and the numerator. Changing only the

reference distribution cannot correct both.

The evidence therefore supports Normal Wald as the general baseline. Expanded Welch is only a conditional sensitivity analysis; future corrections must address the numerator, standard error and population structure together rather than changing the reference law alone.

Appendix A

Additional Derivation Details

This appendix records algebra that supports Chapters 3 and 4 but is not required to follow the main argument.

A.1 Gradient of mutual information on the probability simplex

Mutual information can be written as

$$I(P) = \sum_{i,j} p_{ij} \log p_{ij} - \sum_i p_{i+} \log p_{i+} - \sum_j p_{+j} \log p_{+j}. \quad (\text{A.1})$$

Treating the cell probabilities as coordinates gives

$$\frac{\partial I(P)}{\partial p_{ij}} = (\log p_{ij} + 1) - (\log p_{i+} + 1) - (\log p_{+j} + 1) \quad (\text{A.2})$$

$$= \log \frac{p_{ij}}{p_{i+} p_{+j}} - 1 \quad (\text{A.3})$$

$$= \ell_P(i, j) - 1. \quad (\text{A.4})$$

Any valid probability perturbation $h = (h_{ij})$ satisfies $\sum_{i,j} h_{ij} = 0$. Its directional derivative is therefore

$$\left. \frac{d}{d\varepsilon} I(P + \varepsilon h) \right|_{\varepsilon=0} = \sum_{i,j} \{\ell_P(i, j) - 1\} h_{ij} \quad (\text{A.5})$$

$$= \sum_{i,j} \ell_P(i, j) h_{ij}. \quad (\text{A.6})$$

The constant in the unconstrained gradient disappears on the simplex. For the one-cell direction $h_{ij} = \mathbf{1}\{i = x, j = y\} - p_{ij}$, Equation (A.6) becomes

$$\ell_P(x, y) - \sum_{i,j} p_{ij} \ell_P(i, j) = \ell_P(x, y) - I(P), \quad (\text{A.7})$$

which is Equation (4.19).

A.2 Why the variance of pointwise MI appears

For empirical cell probabilities,

$$\hat{p}_{ij} - p_{ij} = \frac{1}{n_P} \sum_{a=1}^{n_P} \{\mathbf{1}(Z_a^{(P)} = (i, j)) - p_{ij}\}. \quad (\text{A.8})$$

Substitution into the first-order Taylor expansion of MI around P gives

$$\hat{I}(P) - I(P) \approx \sum_{i,j} \ell_P(i, j) (\hat{p}_{ij} - p_{ij}) \quad (\text{A.9})$$

$$= \frac{1}{n_P} \sum_{a=1}^{n_P} \left[\sum_{i,j} \ell_P(i, j) \{\mathbf{1}(Z_a^{(P)} = (i, j)) - p_{ij}\} \right] \quad (\text{A.10})$$

$$= \frac{1}{n_P} \sum_{a=1}^{n_P} \{\ell_P(Z_a^{(P)}) - I(P)\}. \quad (\text{A.11})$$

The observation-level summand has variance

$$\text{Var}_P\{\ell_P(X, Y) - I(P)\} = \mathbb{E}_P \left[\{\ell_P(X, Y) - I(P)\}^2 \right] \quad (\text{A.12})$$

$$= \sum_{i,j} p_{ij} \{\ell_P(i, j) - I(P)\}^2 \quad (\text{A.13})$$

$$= V(P). \quad (\text{A.14})$$

This is why the first-order variance of the complete MI estimator is $V(P)/n_P$, not a variance of the numerical category labels.

A.3 Mean of the variance influence function

The influence function $g_P(x, y)$ is centred at the population level. This can be shown without expanding its individual terms. Write the one-cell direction as

$h^{(x,y)} = \delta_{(x,y)} - P$. Its probability-weighted average is

$$\sum_{x,y} p_{xy} h^{(x,y)} = P - P = 0. \quad (\text{A.15})$$

The first derivative is linear in the direction of change. Therefore, the weighted average of the one-cell derivatives is the derivative in the weighted-average direction:

$$E_P\{g_P(X, Y)\} = \sum_{x,y} p_{xy} \left. \frac{d}{d\varepsilon} V\{(1 - \varepsilon)P + \varepsilon\delta_{(x,y)}\} \right|_0 \quad (\text{A.16})$$

$$= \left. \frac{d}{d\varepsilon} V\left\{P + \varepsilon \sum_{x,y} p_{xy} h^{(x,y)}\right\} \right|_0 \quad (\text{A.17})$$

$$= \left. \frac{d}{d\varepsilon} V(P) \right|_0 = 0. \quad (\text{A.18})$$

The same weighted cancellation holds after exact plug-in substitution. The implementation nevertheless subtracts $\bar{g}_P$ in Equation (4.39) to protect the variance calculation from floating-point roundoff; this does not change the population definition.

A.4 Direct table substitutions

For observed counts $N^P = (N_{ij}^P)$, the quantities used by expanded Welch are calculated as

$$\hat{p}_{ij} = \frac{N_{ij}^P}{n_P}, \quad \hat{p}_{i+} = \sum_j \hat{p}_{ij}, \quad \hat{p}_{+j} = \sum_i \hat{p}_{ij}, \quad (\text{A.19})$$

$$\hat{\ell}_P(i, j) = \log \frac{\hat{p}_{ij}}{\hat{p}_{i+}\hat{p}_{+j}}, \quad \hat{I}(P) = \sum_{i,j} \hat{p}_{ij} \hat{\ell}_P(i, j), \quad (\text{A.20})$$

$$\hat{V}(P) = \sum_{i,j} \hat{p}_{ij} \{\hat{\ell}_P(i, j) - \hat{I}(P)\}^2. \quad (\text{A.21})$$

All sums involving pointwise MI are restricted to cells with $\hat{p}_{ij} > 0$. The probability-weighted row and column means are

$$\hat{R}_P(i) = \frac{\sum_j \hat{p}_{ij} \hat{\ell}_P(i, j)}{\hat{p}_{i+}}, \quad (\text{A.22})$$

$$\hat{C}_P(j) = \frac{\sum_i \hat{p}_{ij} \hat{\ell}_P(i, j)}{\hat{p}_{+j}}. \quad (\text{A.23})$$

Consequently,

$$\hat{g}_P(i, j) = \{\hat{\ell}_P(i, j) - \hat{I}(P)\}^2 - \hat{V}(P) \quad (\text{A.24})$$

$$+ 2\{\hat{\ell}_P(i, j) - \hat{R}_P(i) - \hat{C}_P(j) + \hat{I}(P)\}, \quad (\text{A.25})$$

$$\bar{g}_P = \sum_{i,j} \hat{p}_{ij} \hat{g}_P(i, j), \quad (\text{A.26})$$

$$\hat{\tau}^2(P) = \sum_{i,j} \hat{p}_{ij} \{\hat{g}_P(i, j) - \bar{g}_P\}^2, \quad (\text{A.27})$$

$$\hat{\nu}_P = \frac{2n_P \hat{V}(P)^2}{\hat{\tau}^2(P)}. \quad (\text{A.28})$$

These expressions are the direct $O(rc)$ implementation of the population derivation.

A.5 Numerical derivative checks

The implementation tests the analytic derivatives against centred finite differences on positive probability tables. For a selected cell and a small positive step η , define

$$P_{\pm\eta} = (1 \mp \eta)P \pm \eta\delta_{(x,y)}. \quad (\text{A.29})$$

For sufficiently small η , the checks compare

$$\frac{I(P_\eta) - I(P_{-\eta})}{2\eta} \quad \text{with} \quad \ell_P(x, y) - I(P), \quad (\text{A.30})$$

$$\frac{V(P_\eta) - V(P_{-\eta})}{2\eta} \quad \text{with} \quad g_P(x, y). \quad (\text{A.31})$$

The finite-difference step is selected so that both perturbed tables retain positive probabilities. These checks test the algebra independently of the Monte Carlo calibration experiment.

Appendix B

Population Construction and Exact Design Settings

The experiment uses fixed population tables. This appendix states the construction precisely enough to reproduce any plotted point and records the finite parameter grids. Full-precision probabilities for every constructed pair are stored in `population_definitions.csv`; rounded examples in the thesis are for reading, not simulation input.

B.1 Additive construction

For prescribed positive row vector a and column vector b , form $B = ab^\top$. The principal first-cell direction H contains $\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ in rows and columns 1 and 2 and zero elsewhere. The rare-cell direction places the same block in the last two rows and columns. The spread direction is

$$H_{ij} = u_i v_j, \quad u_i = -1 + \frac{2(i-1)}{r-1}, \quad v_j = -1 + \frac{2(j-1)}{c-1}. \quad (\text{B.1})$$

Both score vectors sum to zero, so every direction preserves margins.

For $t \geq 0$, define $R(t) = B + tH$. Calculate the first strength at which a decreasing cell reaches zero,

$$t_{\max} = \min_{H_{ij} < 0} \frac{B_{ij}}{-H_{ij}}. \quad (\text{B.2})$$

Evaluate the path endpoint MI. A target MI must lie strictly below this endpoint, within the stated numerical tolerance. For a positive target, Brent's one-dimensional root search ([Brent, 1973](#)) solves $I\{R(t)\} = I_{\text{target}}$ on $[0, t_{\max}]$; a zero target uses

$t = 0$. The final table is accepted only if every cell is strictly positive, the sum is one, both margins agree with their prescribed vectors, and achieved MI agrees with the target. The normalisation and margin tolerances are 10^{-12} ; the MI tolerance is $10^{-12} + 10^{-8}I_{\text{target}}$. These numerical checks ensure that a horizontal-axis difference denotes constructed population truth, not an empirical estimate. They do not impose an expected-count floor on the subsequent sample.

The row and column vectors are separately constructed for each population. For shape $r \times c$, profile **uniform** uses uniform vectors of lengths r and c for both P and Q . Profile **same_skew** uses $m_r(0.8)$, $m_c(0.8)$ for both. Profile **different_skew** uses $m_r(0.7)$, $m_c(0.7)$ for P and $m_r(0.8)$, $m_c(0.8)$ for Q . The extreme regimes instead state the two dominant probabilities directly. The direction of the alternative is a separate parameter and should not be inferred from which population has the more skewed margin.

The log-linear comparison uses an ordinal interaction pattern. Its strength is tuned numerically while iterative proportional fitting (Deming and Stephan, 1940) preserves the given row and column margins. The final table is subjected to the same positivity, normalisation, margin and target-MI checks as the additive table. This is a comparison across two ways to make plausible fixed populations, not a claim that a matched target implies matched higher moments or cell counts.

B.2 Frozen statistical grids

Unless a row of Table B.1 says otherwise, both populations use the first-cell additive direction, $I(P) = I_0$, $I(Q) = I_0 + |\Delta|$, and the three standard margin profiles. The symbol n means $n_P = n_Q = n$. A requested point was simulated only after the population constructor verified a strictly positive feasible table. The final display contains no infeasible slots, but that is a result of choosing feasible finite grids, not permission to extend any curve beyond its last constructed target.

Table B.1: Exact statistical experiment grids; MI values are in nats.

Family	Settings
Main	Shapes $2 \times 2, 3 \times 3, 5 \times 5, 8 \times 8$; three profiles; $I_0 = 0.02$; $ \Delta \in \{0, 0.001, 0.002, 0.005, 0.01, 0.02\}$; $n \in \{2, 5, 10, 20, 50, 100, 250, 500, 1000\}$.
Baseline	Main shapes and profiles; $I_0 \in \{0.0001, 0.001, 0.02\}$; main difference grid; $n \in \{20, 100, 1000\}$.

Family	Settings
Changed cells	Shapes $3 \times 3, 5 \times 5, 8 \times 8$; different-skew margins; $I_0 = 0.0001$; $ \Delta \in \{0, 0.0001, 0.00025, 0.0005, 0.001, 0.002\}$; $n \in \{20, 100, 1000\}$. First and rare blocks for all shapes; spread pattern for 5×5 and 8×8 .
Imbalance	Main shapes, profiles and difference grid; $I_0 = 0.02$. Sample pairs are $(50, 50), (50, 100), (100, 50), (50, 250), (250, 50), (50, 500), (500, 50), (500, 500)$. Both Q -higher and P -higher MI directions.
Larger effects	Shapes $2 \times 2, 3 \times 3$; three profiles; $I_0 = 0.02$; main difference grid plus $0.05, 0.1, 0.2$; $n \in \{20, 100, 1000\}$.
Rectangular	Shapes $2 \times 3, 3 \times 5$; three profiles; $I_0 = 0.02$; main difference grid; $n \in \{20, 100, 1000\}$.
Construction	Shapes $3 \times 3, 5 \times 5, 8 \times 8$; uniform and different-skew profiles; $I_0 \in \{0.0001, 0.02\}$; main difference grid; $n \in \{20, 100, 1000\}$; additive first block and ordinal log-linear constructors.
Extreme skew	Shapes $2 \times 2, 3 \times 3, 8 \times 8$; dominant margin probability pairs $(\rho_P, \rho_Q) \in \{(0.9, 0.95), (0.99, 0.995), (0.999, 0.9995)\}$; $I_0 = 0.00001$; $ \Delta \in \{0, 0.00001, 0.00005, 0.0001, 0.0002\}$; $n \in \{1, 2, 5, 20, 100, 1000, 10000\}$.
Rare-cell stress	Shapes $3 \times 3, 8 \times 8$; dominant pair $(0.9, 0.95)$; first and rare blocks; $I_0 = 0.00001$; $ \Delta \in \{0, 0.00001, 0.000025, 0.00005, 0.0001\}$; same n ladder as extreme skew.
Independence	Shapes $2 \times 2, 8 \times 8$; three profiles; $I_0 = 0$; main difference grid; $n \in \{2, 20, 100, 1000\}$.
Convergence	Main square shapes and profiles; equal MI $I(P) = I(Q) \in \{0.0001, 0.02\}$; $n \in \{1000, 2500, 10000, 50000\}$. Additional rare-block controls use $3 \times 3, 5 \times 5, 8 \times 8$, different-skew margins and common MI 0.0001 .

The $n = 1$ stress setting is retained and not hidden; the statistic is not evaluable there and its result is recorded as invalid. Every unique statistical configuration has 20,000 sampled table pairs. When the same population pair and sample sizes appear in another section, the saved configuration is reused for presentation and not simulated again.

Appendix C

Evidence Index and Additional Diagnostics

The main results chapter shows a systematic selection of full-scale and calibration-zoom figures. The full companion atlas remains the source for every sample size and focused regime in the frozen design. The index below specifies how to retrieve exact numbers without averaging regimes into a single summary. Paths are relative to the workspace root `Simulations/`.

The companion written atlas is at `WelchSatterthwaiteMI/docs/experiments/THESIS_EXPERIMENTS.md`. Its 99 primary figures and seven extra calibration zooms retain the full design. The thesis-specific PDFs in `Thesis Writeup/Welch MI/figures_rewrite/` are generated from the two final CSV files named in the source script, not from the old pilot folders. The adjacent `figure_manifest.json` lists, for each thesis PDF, its original section, source files, configuration identifiers and 20,000-replicate denominator. The file `evidence_values.tex` is generated from exact configuration IDs for the numerical examples in Chapter 6.

The plotted rate is $R^{-1} \sum_s A_{m,s}$, where $A_{m,s}$ requires both a valid output and rejection. The method-valid conditional rate divides valid rejections by that method's valid outputs; it is undefined if there are none. The common-valid rate restricts to outputs where both methods are valid. A paired difference is based on counts of the four joint rejection outcomes for the same table pairs. These distinctions allow a reader to check whether a curve difference is primarily a changed reference or lost validity.

Specifically, let $A_{m,s}$ be the valid-rejection indicator for method m on replicate s , and set $D_s = A_{\text{expanded},s} - A_{\text{Wald},s}$. The saved paired difference is $\bar{D} = R^{-1} \sum_s D_s$,

Table C.1: Saved files and the question each answers.

File in the final results directory	Use
<code>display_manifest.csv</code>	Figure section, row, column, horizontal value and exact configuration identifier; a configuration may recur.
<code>configuration_manifest.csv</code>	Unique statistical configurations, sample sizes, target populations and simulation identifiers.
<code>population_definitions.csv</code>	Full-precision P and Q probability tables, margins, achieved MI, constructor and feasibility diagnostics for every pair.
<code>cell_results.csv</code>	One row per configuration and method: unconditional rejection, method validity, conditional and common-valid rates, Wilson intervals, expected-count and support-loss diagnostics.
<code>paired_method_results.csv</code>	Paired rejection counts, rate differences and approximate paired intervals from the same replicates.
<code>runtime_summary.csv</code>	Per-regime median and interquartile call times in microseconds, validity and Expanded-to-Wald time ratio.
<code>verification.json</code> and <code>report_verification.json</code>	Run-level population, denominator, pairing, method and report checks.

with approximate 95% interval

$$\bar{D} \pm 1.96 \left\{ \frac{1}{R(R-1)} \sum_{s=1}^R (D_s - \bar{D})^2 \right\}^{1/2}. \quad (\text{C.1})$$

This calculation preserves the within-replicate pairing. The individual method intervals use Wilson’s binomial score construction ([Wilson, 1927](#)).

For any P population with sample size n_P , its minimum expected cell count is $\min_{ij} n_{PPij}$; define Q ’s analogously. No “expected minimum” of random observed counts is calculated. The saved support diagnostics include empty observed rows and columns and the fraction of zero-count cells. Such values describe why a statistic might fail, but do not constitute a universal validity threshold.

The thesis does not treat a descriptive count of regimes closer to 0.05 as a formal paired test over regimes. Wilson intervals quantify binomial Monte Carlo variation at each fixed point, and paired normal intervals quantify a specified within-configuration method difference. Neither interval family is a simultaneous assertion across all configurations. Tables sharing a population but changing n or $|\Delta|$ answer different fixed-design questions and should not be collapsed into one mean rate.

Appendix D

The Independence Boundary

The main thesis concerns the regular weak null $I(P) = I(Q) > 0$. This appendix explains why constructing an independent reference distribution does not turn the Expanded Welch method into a new first-order test of independence.

D.1 Constructing the independent reference

For a joint distribution $P = (p_{ij})$, define

$$q_{ij} = p_{i+}p_{+j}. \quad (\text{D.1})$$

Then $Q = (q_{ij})$ is a probability distribution because

$$\sum_{i,j} q_{ij} = \left(\sum_i p_{i+} \right) \left(\sum_j p_{+j} \right) = 1. \quad (\text{D.2})$$

It has the same margins as P , since

$$\sum_j q_{ij} = p_{i+}, \quad \sum_i q_{ij} = p_{+j}. \quad (\text{D.3})$$

The factorisation in Equation (D.1) makes X and Y independent under Q , so $I(Q) = 0$.

The divergence from P to this reference is

$$D_{\text{KL}}(P \| Q) = \sum_{i,j} p_{ij} \log \frac{p_{ij}}{q_{ij}} \quad (\text{D.4})$$

$$= \sum_{i,j} p_{ij} \log \frac{p_{ij}}{p_{i+}p_{+j}} \quad (\text{D.5})$$

$$= I(P). \quad (\text{D.6})$$

Consequently, testing $I(P) = I(Q)$ for this constructed reference is exactly testing $I(P) = 0$, or independence.

D.2 Why the first-order term vanishes

At independence, $p_{ij} = p_{i+}p_{+j}$, and every population pointwise-MI value is zero:

$$\ell_P(i, j) = \log 1 = 0. \quad (\text{D.7})$$

Equation (4.19) therefore gives

$$\left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0 = \ell_P(x, y) - I(P) = 0. \quad (\text{D.8})$$

The first-order variance also vanishes:

$$V(P) = \text{Var}_P\{\ell_P(X, Y)\} = 0. \quad (\text{D.9})$$

The denominator used by Normal Wald and Expanded Welch is thus degenerate under the population independence null. This does not mean that a finite sample must have zero plug-in MI or zero plug-in variance. Sampling noise usually makes the empirical table depart from exact independence. The point is that these random departures do not enter MI through the usual first-order term at the null. They enter through the second-order term derived next, so changing only the degrees of freedom of a first-order statistic cannot supply the missing source of variation.

D.3 Second-order behaviour

Fix a strictly positive independent table P , and let $h = (h_{ij})$ describe a small redistribution of probability, with $\sum_{i,j} h_{ij} = 0$. Define the path

$$P_\varepsilon = P + \varepsilon h, \quad f(\varepsilon) = I(P_\varepsilon), \quad (\text{D.10})$$

for values of ε small enough that all cells remain positive. Here $\varepsilon = 0$ is the expansion point and $f(0) = I(P) = 0$. The second-order Taylor expansion is

$$f(\varepsilon) \approx \underbrace{f(0)}_{\text{zeroth order}} + \underbrace{f'(0)\varepsilon}_{\text{first order}} + \underbrace{\frac{f''(0)}{2!}\varepsilon^2}_{\text{second order}}. \quad (\text{D.11})$$

The first-order term is

$$f'(0) = \sum_{i,j} \{\ell_P(i,j) - 1\} h_{ij} = 0. \quad (\text{D.12})$$

Both parts vanish: $\ell_P(i,j) = 0$ at independence, and the probability constraint gives $\sum_{i,j} h_{ij} = 0$. Thus, positive and negative cell errors cannot create a linear change in MI around the independence table.

To state the surviving term, define the row and column changes $h_{i+} = \sum_j h_{ij}$ and $h_{+j} = \sum_i h_{ij}$. Differentiating the entropy form of MI twice gives

$$f''(0) = \sum_{i,j} \frac{h_{ij}^2}{p_{ij}} - \sum_i \frac{h_{i+}^2}{p_{i+}} - \sum_j \frac{h_{+j}^2}{p_{+j}}. \quad (\text{D.13})$$

Because $p_{ij} = p_{i+}p_{+j}$, define the part of each cell change that cannot be explained by changed margins as

$$a_{ij} = h_{ij} - p_{+j}h_{i+} - p_{i+}h_{+j}. \quad (\text{D.14})$$

Equation (D.13) can then be written as

$$f''(0) = \sum_{i,j} \frac{a_{ij}^2}{p_{i+}p_{+j}}. \quad (\text{D.15})$$

Substituting the zeroth-, first-, and second-order terms into Equation (D.11) gives

$$I(P_\varepsilon) = \underbrace{0}_{\text{zeroth and first order}} + \underbrace{\frac{\varepsilon^2}{2} \sum_{i,j} \frac{a_{ij}^2}{p_{i+}p_{+j}}}_{\text{second order}} + o(\varepsilon^2). \quad (\text{D.16})$$

The remainder $o(\varepsilon^2)$ is negligible relative to ε^2 as ε approaches zero.

For a multinomial sample, take $\varepsilon h = \hat{P} - P$. An empirical cell proportion has standard deviation $\sqrt{p_{ij}(1-p_{ij})/n}$, so each εh_{ij} is typically proportional to $n^{-1/2}$. The corresponding row- and column-adjusted errors εa_{ij} have the same scale. Equation (D.16) squares these errors, making $\hat{I}(P)$ proportional to n^{-1} under independence. Multiplying by $2n$ therefore produces a statistic with a non-vanishing scale:

$$2n\hat{I}(P) \xrightarrow{d} \chi_{(r-1)(c-1)}^2. \quad (\text{D.17})$$

This is the usual likelihood-ratio, or G , test of independence under a fixed alphabet and the standard positive-support large-sample conditions (Wilks, 1938; Brillinger, 2004). The chi-squared reference describes the squared cell errors in Equation (D.16).

Expanded Welch instead standardises a signed first-order MI error on the $n^{-1/2}$ scale. At independence that signed term is absent, so its Student reference is not the chi-squared reference in another form.

The same likelihood-ratio connection appears for conditional information measures. For finite Markov chains, [Barnett and Bossomaier \(2012\)](#) relate transfer entropy to a log-likelihood ratio with an asymptotic chi-squared reference; the wider information-theoretic setting is reviewed by [Bossomaier et al. \(2016\)](#). This is an adjacent application of the same second-order likelihood principle, not a two-sample equal-MI Welch test.

D.4 What alternative constructions produce

Using the observed margins to set $\hat{q}_{ij} = \hat{p}_{i+}\hat{p}_{+j}$ gives

$$2nD_{\text{KL}}(\hat{P}\|\hat{Q}) = 2n\hat{I}(P), \quad (\text{D.18})$$

which is the likelihood-ratio statistic above. Because $\hat{Q}$ is constructed from the same sample as $\hat{P}$, it is not an independent second table and cannot be inserted into the two-sample Welch variance formula as though it supplied new information.

Repeatedly shuffling one variable preserves its observations while breaking the pairing with the other variable. The resulting empirical distribution is a permutation reference. Repeatedly drawing tables from $\hat{P}_X\hat{P}_Y$ is a parametric bootstrap reference. A single simulated Q table adds Monte Carlo variation but does not by itself define a calibrated null distribution.

The independent reference $Q = P_X P_Y$ is therefore mathematically valid, but it does not create a regular two-sample MI comparison. Analytic improvement of the sparse independence test would require a second-order finite-sample method and is not an immediate extension of the Expanded Welch calculation.

Appendix E

Reproducibility and Protocol Record

The final protocol is `WelchSatterthwaiteMI/experiments/THESIS_REDESIGN_PROTOCOL.json`, version 1.0.0, frozen on 7 September 2026 UTC before the final simulation. It specifies the population grids, seeds, methods, 20,000 replicates, $\alpha = 0.05$, validation tolerances and timing design. This was a local protocol freeze after exploratory development, not an external preregistration.

The run contains 3,111 unique configurations, 4,001 display positions and 534 population pairs, corresponding to 62,220,000 sampled table pairs. The saved verification records check the population targets, result denominators, paired method outputs, validity rules and generated atlas. Chapter 6 uses the same frozen tables; no population was resimulated for a thesis figure.

The exact software versions, seeds and source hashes are recorded in `WelchSatterthwaiteMI/results/thesis_redesign/run_metadata.json`. The accompanying experiment supplement contains the protocol, implementation, consolidated results, 3,111 configuration checkpoints, complete atlas, mechanism diagnostics and tests. Its README also documents the minor post-run changes to the orchestration script; the population construction, simulation and method calculations used for the frozen results are unchanged.

The thesis evidence audit reconstructs the protocol manifests, identifies the source configurations for every displayed figure and checks every numerical claim in Chapter 6 against the saved files. The final deliverable checksums are recorded in `SHA256SUMS.txt`. After extracting the source and supplement together, run `python figures_rewrite/make_figures.py`, `python figures_rewrite/audit_evidence.py` and `latexmk -pdf main.tex`. Runtime definitions and limitations are given with the results in Section 6.8.

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